

Name: _____

Double Award Applied Science – Unit 3 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /75	Grade	Good Questions	Bad Questions	Reasons
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	G	F	E	D	C	Max UMS
2018	10	14	18	23	28	33
2019	13	17	21	25	30	35
2022	12	16	20	25	30	36
2023	14	18	22	27	32	37

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – NEW

3445U30-1



APPLIED SCIENCE (Double Award)
UNIT 3: Food, Materials and Processes

FOUNDATION TIER

TUESDAY, 15 MAY 2018 – AFTERNOON

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	13	
3.	6	
4.	15	
5.	12	
6.	14	
7.	5	
Total	75	

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ADDITIONAL MATERIALS

A calculator.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **5(a)(i)** is a quality of extended response (QER) question where your writing skills will be assessed.
The Periodic Table is printed on the back cover of the examination paper.

*Answer **all** questions in the spaces provided.*

1. There are different types of material in use today. Their uses are determined by their properties.

- (a) (i) Complete the table below by selecting the word from the box to identify the type of material from its description.

Each word may be used once, more than once or not at all.

[3]

polymer	ceramic	composite	alloy	metal
---------	---------	-----------	-------	-------

Description	Type of material
hard, brittle, high resistance	
hard, high melting point, good conductor	
mixture of two or more elements	

- (ii) The properties of materials depend on their structure.
Describe the structure of polymers.

[2]

.....

.....

.....

- (b) Metals are very useful materials in a variety of industries. One use is for making wings of modern aircraft.

Some properties of the metals that could be used are given in the table below.

Metal	Density (kg/m ³)	Stiffness (GPa)	Tensile strength ($\times 10^7$ Pa)
titanium	4 500	110	55
aluminium	2 700	69	10
vanadium	5 700	138	63
steel	7 800	210	40

Use the information from the table above to answer the following questions.

- (i) State which metal could be used to make the wings as flexible as possible. [1]

.....

- (ii) I State which metal could be used to make the wings as light as possible. [1]

.....

- II Give **one** reason for your answer. [1]

.....

- (iii) I State which metal could be used so the wings can withstand large forces. [1]

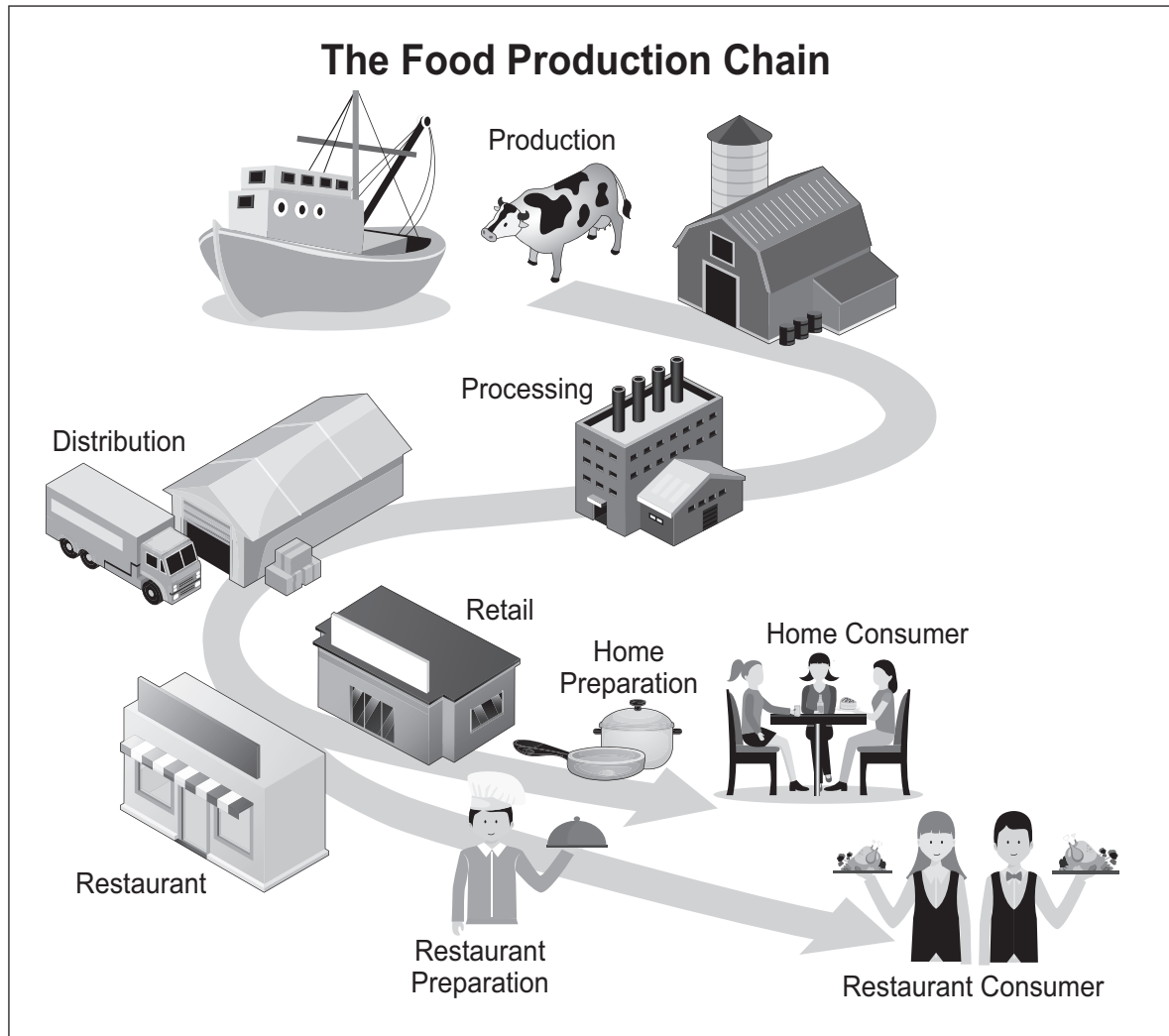
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- II Give **one** reason for your answer. [1]

.....

2. The food industry uses fruits, vegetables, grains and raw meat and processes these into the food products that are sold in supermarkets. Food production often involves cleaning and cooking raw foods, as well as the addition of additives and other ingredients.

One purpose of food production processes is to create products with longer shelf-lives than raw food ingredients.



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The food industry must take precautions to prevent consumers becoming ill from food poisoning.

- (a) Complete the table below which shows some of the methods of controlling bacteria in the food industry. [3]

Method	Effect on bacteria
refrigeration	slows down bacterial growth
freezing	
.....	kills bacteria
.....	dries bacteria to prevent growth

(b) Sometimes food becomes contaminated with bacteria, causing food poisoning.

(i) State **one** symptom of food poisoning.

[1]

.....

(ii) Name **one** type of bacteria that causes food poisoning.

[1]

.....

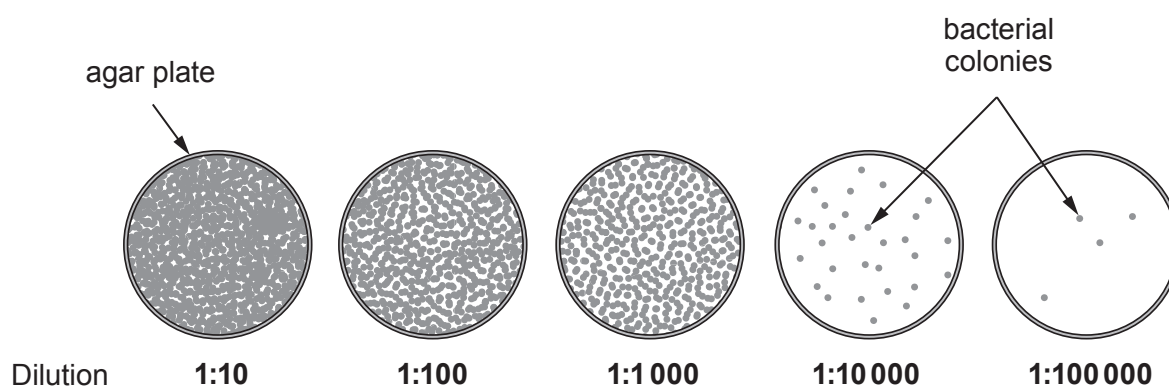
(iii) State the harmful substances produced by bacteria as they grow.

[1]

.....

(c) Whenever an outbreak of food poisoning occurs, environmental health officers test samples of food for the presence of bacteria.

(i) To be able to count the number of bacterial colonies a series of dilutions is carried out as shown in the diagram below.



The 1:10 000 dilution was used to calculate the number of bacteria present.

I Explain why using a dilution of 1:10 will not provide valid results.

[2]

.....

.....

.....

II Give **one** reason why the 1:100 000 dilution was also not used to calculate the number of bacteria present.

[1]

.....

- (ii) In an investigation, six agar plates were prepared by adding 1 cm^3 of solution from a 1:10 000 dilution. The agar plates were incubated for 72 hours. The results are shown below.

Plate number	1	2	3	4	5	6
Number of colonies	35	35	25	15	30	25

- I State which plate has produced an anomalous result.

[1]

.....

- II Use the equation:

$$\text{number of bacteria/cm}^3 = \text{mean number of colonies} \times 10\,000$$

to calculate the number of bacteria/cm³ in the original sample.

[3]

Number of bacteria/cm³ =

13

3. Forensic scientists analyse substances found at crime scenes. They routinely identify powder samples.

Students are using the same techniques to analyse an unknown powder.

- (a) The students dissolve the unknown powder in water and add sodium hydroxide solution (NaOH) to test for metal cations. They then add excess NaOH solution to see if the precipitate dissolves or not.

The table shows results for a selection of cations.

Cation	Colour of precipitate when a little NaOH solution is added	Precipitate dissolves when excess NaOH solution is added (✓ or ×)
Ca^{2+}	white	×
Zn^{2+}	white	✓
Pb^{2+}	white	✓
Mg^{2+}	white	×
Cu^{2+}	blue	×
Fe^{2+}	green	×

They noticed that a white precipitate was produced which did not dissolve in excess NaOH solution.

- (i) State the symbols for the **two** metal cations that could be present. [1]

.....

- (ii) The table below shows the flame tests for a selection of metal cations.

Metal cation	Flame colour
Ca^{2+}	brick-red
Zn^{2+}	greyish-white
Pb^{2+}	greyish-white
Mg^{2+}	no colour
Cu^{2+}	blue-green
Fe^{2+}	gold

- Describe how a flame test could be used to confirm which cation is present. [2]

.....
.....

- (b) They use another sample of the powder and perform the following tests to identify the anions that may be present.

Anion	Test	Observation
bromide (Br^-)	add aqueous silver nitrate	cream precipitate
carbonate (CO_3^{2-})	add hydrochloric acid	bubbles of gas
chloride (Cl^-)	add aqueous silver nitrate	white precipitate
iodide (I^-)	add aqueous silver nitrate	yellow precipitate
sulfate (SO_4^{2-})	add aqueous barium chloride	white precipitate

The observations from the students' tests are:

- bubbles of gas are given off when hydrochloric acid is added
- a white precipitate forms when aqueous silver nitrate is added

- Use the information above to determine the **two** anions present in the powder. [2]

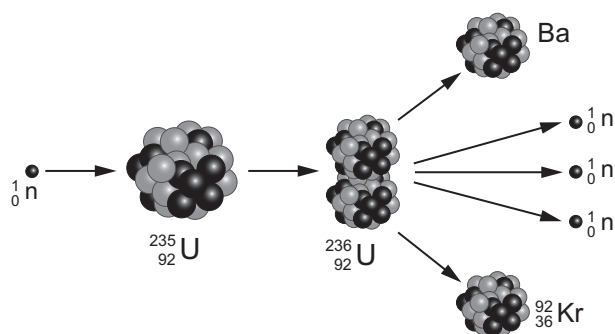
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- (c) Name **one** compound that could be present in the powder. [1]

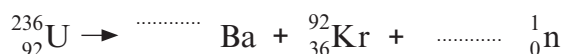
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4. During their operation nuclear reactors produce large quantities of electrical energy without creating greenhouse gases.

(a) One type of reaction involving uranium-235 is shown in the diagram below.



- (i) Use the information in the diagram to complete the balanced symbol equation for part of this reaction. [3]



- (ii) Underline the correct word(s) in the brackets in the paragraph below. [4]

The absorption of slow neutrons can cause a (**fusion** / **fission** / **chemical**) reaction in uranium nuclei.

The neutrons are slowed down by (a **moderator** / **control rods** / **concrete shielding**).

The emission of (**protons** / **neutrons** / **electrons**) in this reaction can cause a chain reaction.

An uncontrolled chain reaction is prevented by using (a **moderator** / **control rods** / **concrete shielding**).

- (iii) Complete the table below about the two uranium isotopes. [2]

Uranium isotope	Number of protons	Number of neutrons
uranium-235		143
uranium-236		

- (b) Information about two natural isotopes of uranium is given below.

Isotope	Symbol	Type of decay	Half-life (years)
uranium-235	${}_{92}^{235}\text{U}$	alpha	700 million
uranium-238	${}_{92}^{238}\text{U}$	alpha	4.5 billion

- (i) Put ticks (✓) in the **two** boxes that correctly give the meaning of the half-life of a radioactive substance. [2]

Half the time for the radioactivity to halve

☐

The time taken for an atom to split in half

☐

The time taken for the number of undecayed nuclei to halve

☐

The time taken for the activity to halve

☐

The time taken for half of the alpha particles to decay

☐

- (ii) The isotopes of uranium in the table have been in existence since the Earth was formed. One estimate of the Earth's age is 4.5 billion years. Calculate the fraction of uranium-238 that remains on Earth today. [2]

Fraction =

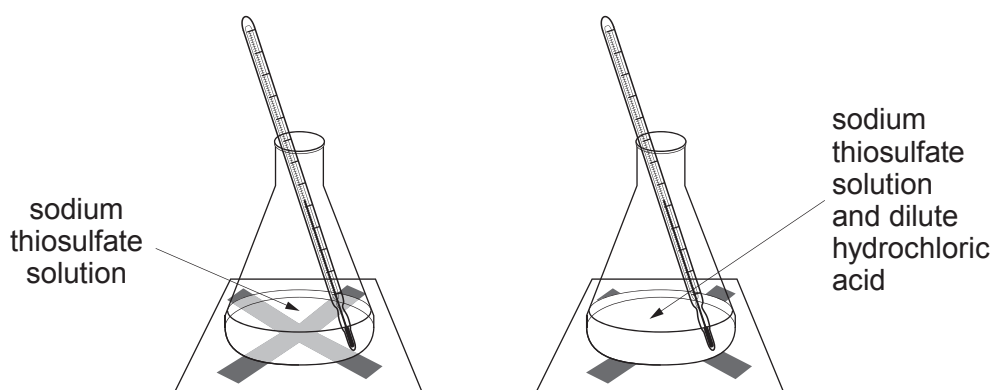
- (iii) When the Earth was formed, it contained equal amounts of uranium-235 and uranium-238. It is suggested that more uranium-235 remains today than uranium-238. Explain whether or not you agree with this suggestion. [2]

.....

.....

.....

5. The chemical industry is developing new processes to manufacture chemicals more efficiently.
- (a) (i) When sodium thiosulfate reacts with hydrochloric acid the solution becomes cloudy.



Describe a method used to investigate how temperature affects the rate of the reaction between hydrochloric acid and sodium thiosulfate solution. Include details of apparatus and volumes of chemicals used in your answer. [6 QER]

(ii) State the **two** variables that need to be controlled in this experiment. [2]

1

2

(iii) State **one** reason why these variables need to be controlled. [1]

.....
.....

(b) In a different experiment hydrochloric acid was poured into a flask containing different volumes of sodium thiosulfate solution and water. The time taken to react was measured. The results are shown in the table below.

Volume of acid (cm ³)	Volume of sodium thiosulfate (cm ³)	Volume of water (cm ³)	Time for cross to disappear from view (s)
60	55	5	44
60	50	10	51
60	45	15	57
60	40	20	83
60	35	25	106

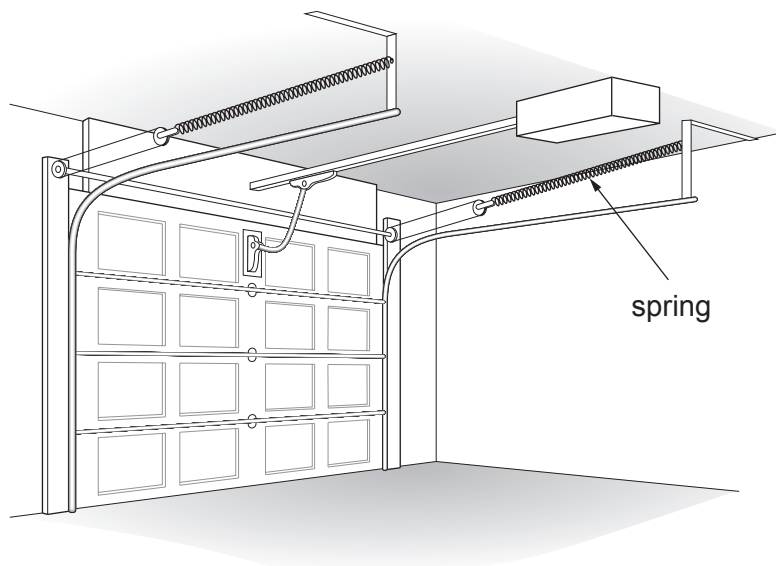
(i) State what conclusion can be made from the results. [1]

.....
.....

(ii) Explain the results in terms of particles. [2]

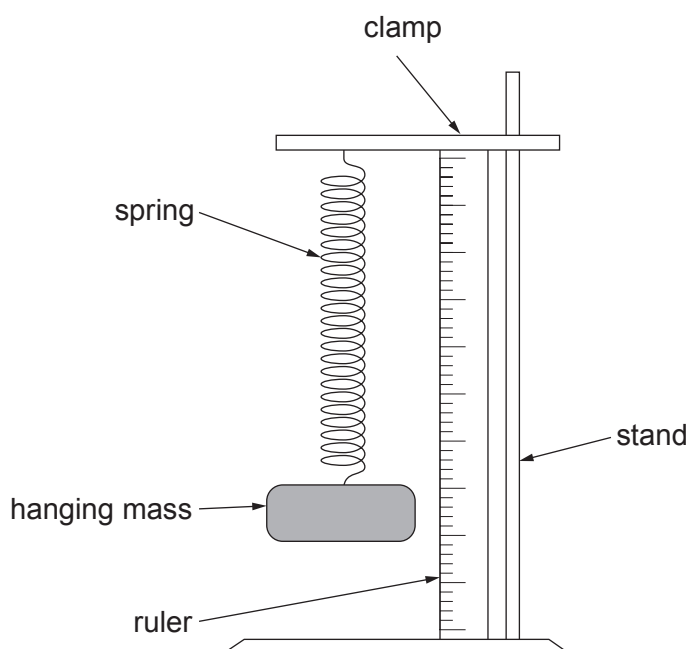
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6. Springs can be used in garage doors. When the door shuts the springs are stretched and put under tension. When the door opens the spring returns to its original length making it easier to raise.

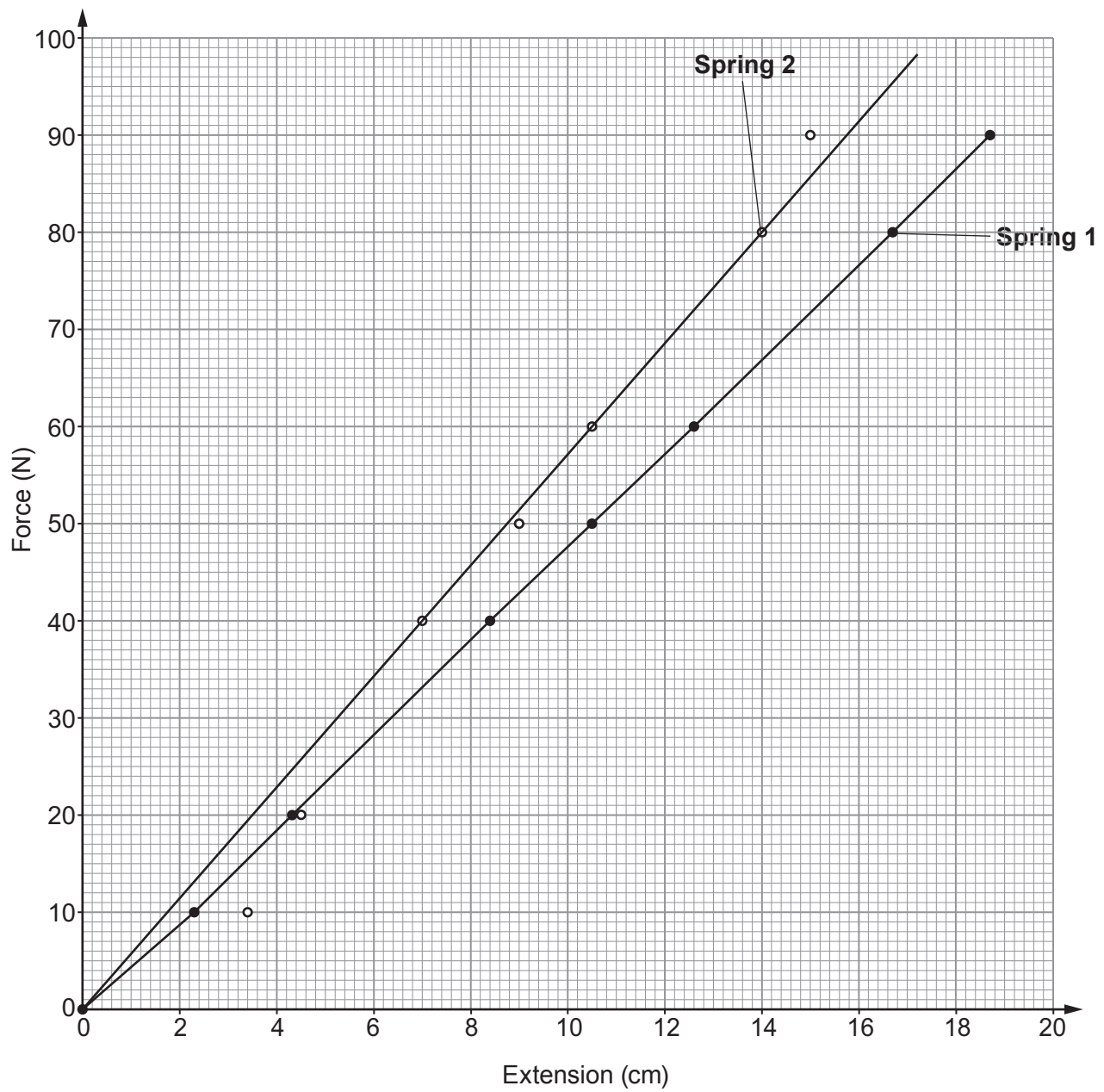


It is important that the spring is designed so that it does not become permanently stretched during the process. Springs become permanently stretched once they pass their **elastic limit**. This is the point where extension is no longer proportional to the stretching force.

Students test three springs using the apparatus shown below.



(a) Their results for two springs used in the doors are shown in the graph below.



- (i) Compare the behaviour of the springs as the force increases from 10 N to 40 N.

[2]

.....

.....

.....

- (ii) Use the graph and the equation:

$$\text{spring constant} = \frac{\text{force}}{\text{extension}}$$

to calculate the value of the spring constant for **Spring 1**.

[3]

spring constant = N/cm

- (iii) Use the graph to state **one** reason why the students can have more confidence in the results for **Spring 1** than for **Spring 2**.

[1]

.....

.....

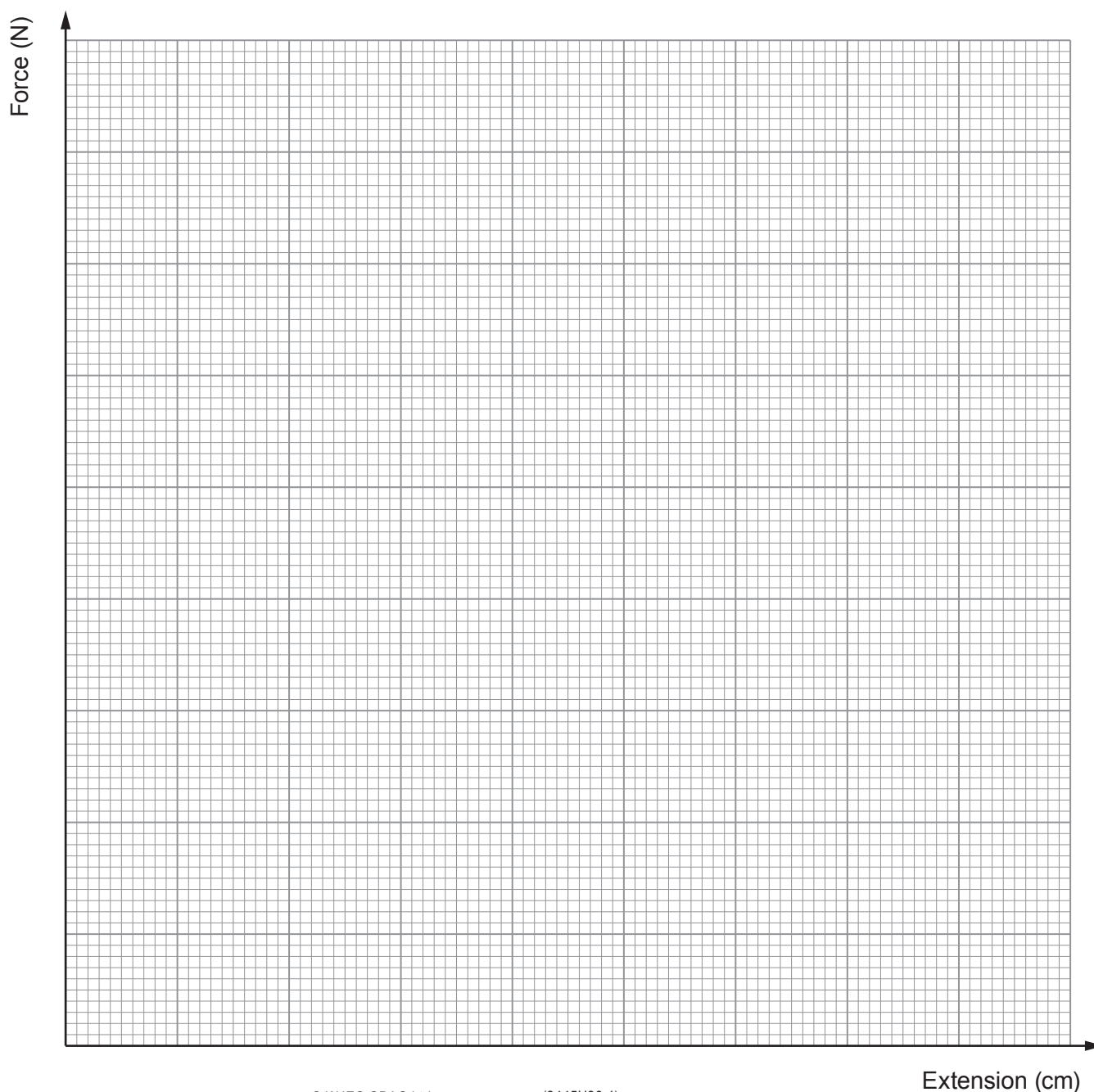
(b) The results for **Spring 3** are given in the table below.

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Force (N)	Extension (cm)
0	0.0
10	2.4
20	4.8
40	9.6
50	12.0
60	16.0
80	28.0
90	36.0

(i) Plot the data on the grid below and draw a suitable line.

[4]



- (ii) Springs become permanently stretched once they pass their **elastic limit**. This is the point where extension is no longer proportional to the stretching force.

I. **Label** the elastic limit on your graph with a letter **E**. [1]

II. Use your graph to find the force required to reach the elastic limit. [1]

Force = N

- (iii) During the opening and closing of the garage doors the spring will extend by 30 cm. Explain whether or not **Spring 3** is suitable for this purpose. [2]

.....

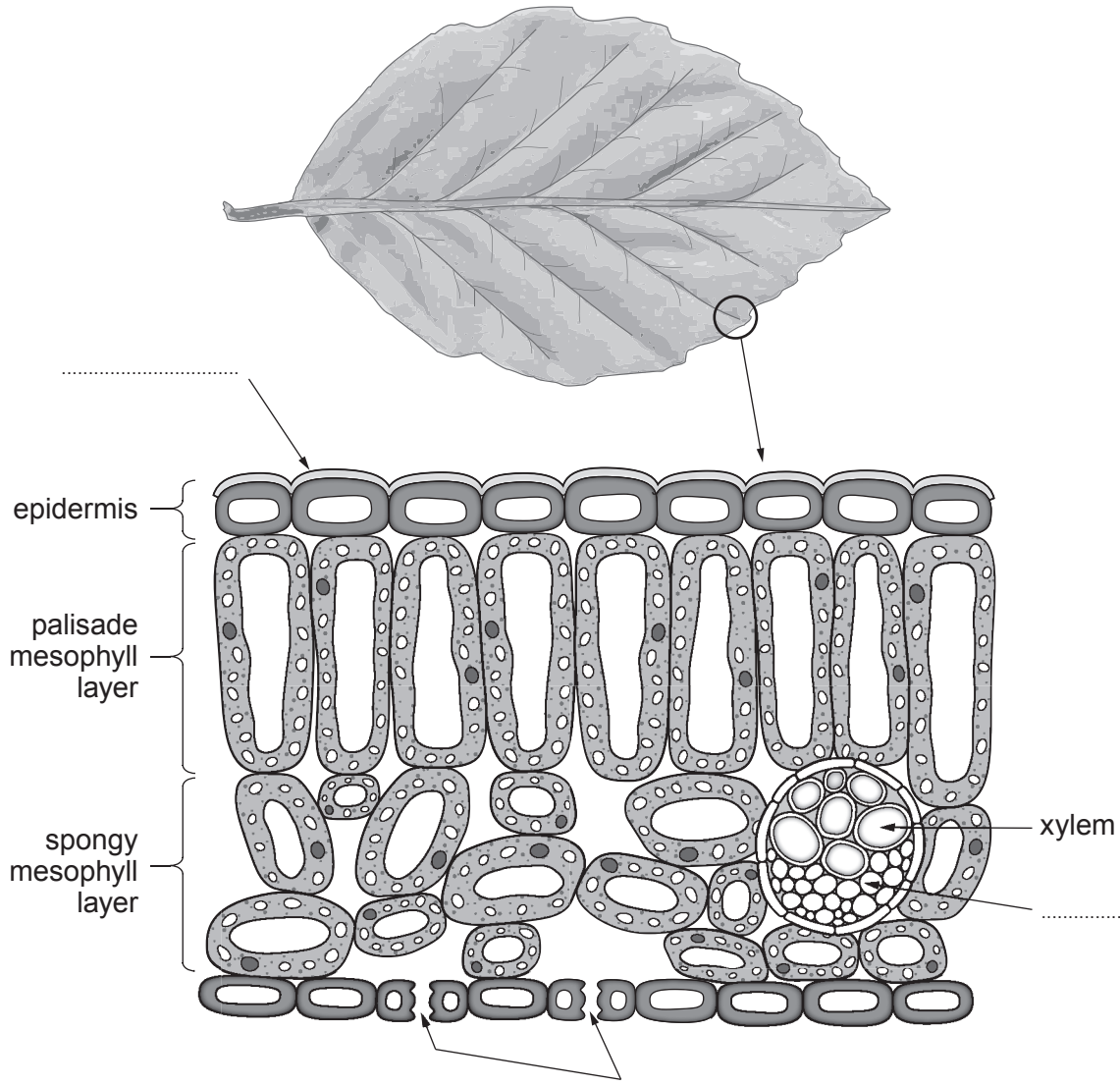
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7. Students are learning about plants. They are interested in the structure and function of leaves.

(a) **Complete** the labelling of the parts of the leaf in the diagram below.

[3]



(b) State the advantage of a transparent epidermis.

[1]

.....

.....

(c) State the purpose of the spongy mesophyll layer containing air spaces.

[1]

.....

.....

END OF PAPER

THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																				4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											19 F Fluorine 9	20 Ne Neon 10							
23 Na Sodium 11	24 Mg Magnesium 12											32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18						
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36			
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54				
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86			
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																		

Key

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number

Surname	Centre Number	Candidate Number
Other Names		0



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TUESDAY, 14 MAY 2019 – AFTERNOON

APPLIED SCIENCE (Double Award)
UNIT 3: Food, Materials and Processes

FOUNDATION TIER

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	11	
3.	7	
4.	6	
5.	6	
6.	10	
7.	7	
8.	7	
9.	12	
Total	75	

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ADDITIONAL MATERIALS

A calculator.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **5** is a quality of extended response (QER) question where your writing skills will be assessed.
The Periodic Table is printed on the back cover of the examination paper.

Answer **all** questions in the spaces provided.

1. Drinks manufacturers use standard food colourings which are identified by E-numbers. A student was asked to investigate which E-numbers were present in different brands of orange drinks.

The student suggested the following method:

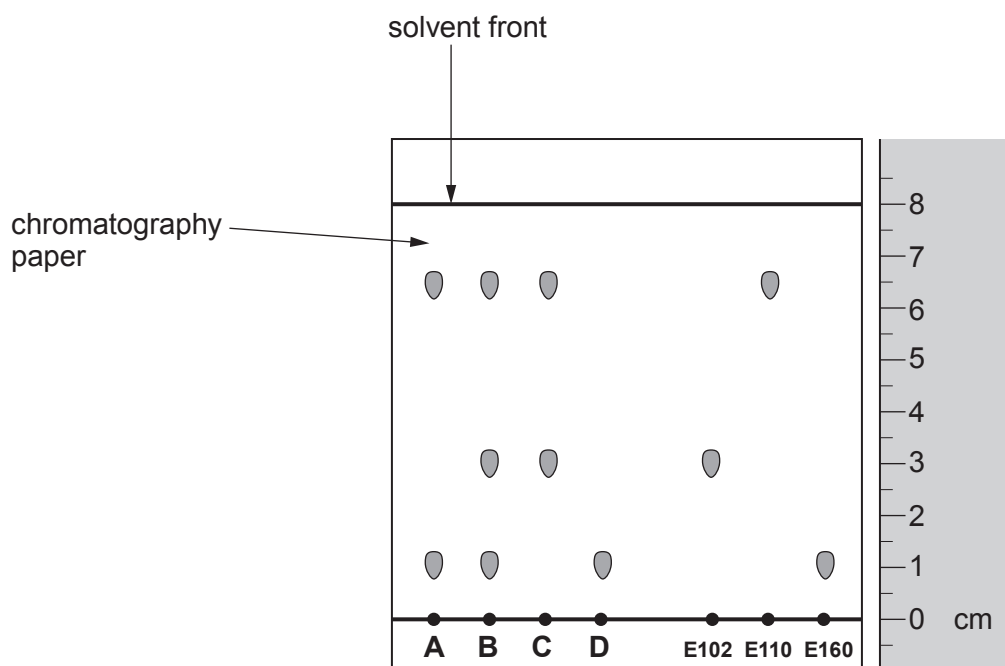
1. Place spots of each drink **A**, **B**, **C** and **D** on a pen line on chromatography paper.
2. Place spots of three common food colourings, **E102**, **E110**, **E160** on the pen line.
3. Hang the chromatography paper in a beaker and add water until it is above the pen line.
4. Leave the paper to stand in water in a beaker until the water rises to near the top of the chromatography paper.

- (a) The student failed to produce a chromatogram using this method. State **two** errors in this method. [2]

1.

2.

- (b) Another student, using a valid method, obtained the chromatogram below (*not drawn to scale*).



Answer the following questions using the information in the chromatogram.

- (i) State which food colouring (**E102**, **E110**, **E160**) is the most soluble. [1]

.....

- (ii) State which drinks (**A**, **B**, **C** or **D**) contained **E110**. [1]

.....

- (iii) The student predicted that all orange drinks contain the same E-numbers. Explain whether this prediction was correct by comparing the food colourings found in drinks **B** and **D**. [2]

.....

.....

.....

- (iv) The retention factor (R_f) value of a substance can be used to identify that substance from a chromatogram.

- I. State the distance travelled by the solvent front. [1]

Distance = cm

- II. Use the equation:

$$R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by the solvent front}}$$

- to calculate the R_f value for **E102**. [2]

R_f =

2. Hydrogen peroxide, H_2O_2 , has anti-fungal properties and can be used to preserve milk. Before the milk is used to make cheese the H_2O_2 needs to be removed.

Hydrogen peroxide decomposes to form water and oxygen gas as shown below.

hydrogen peroxide $\rightarrow$ water + oxygen

- (a) Complete the balanced **symbol** equation for the reaction. [2]

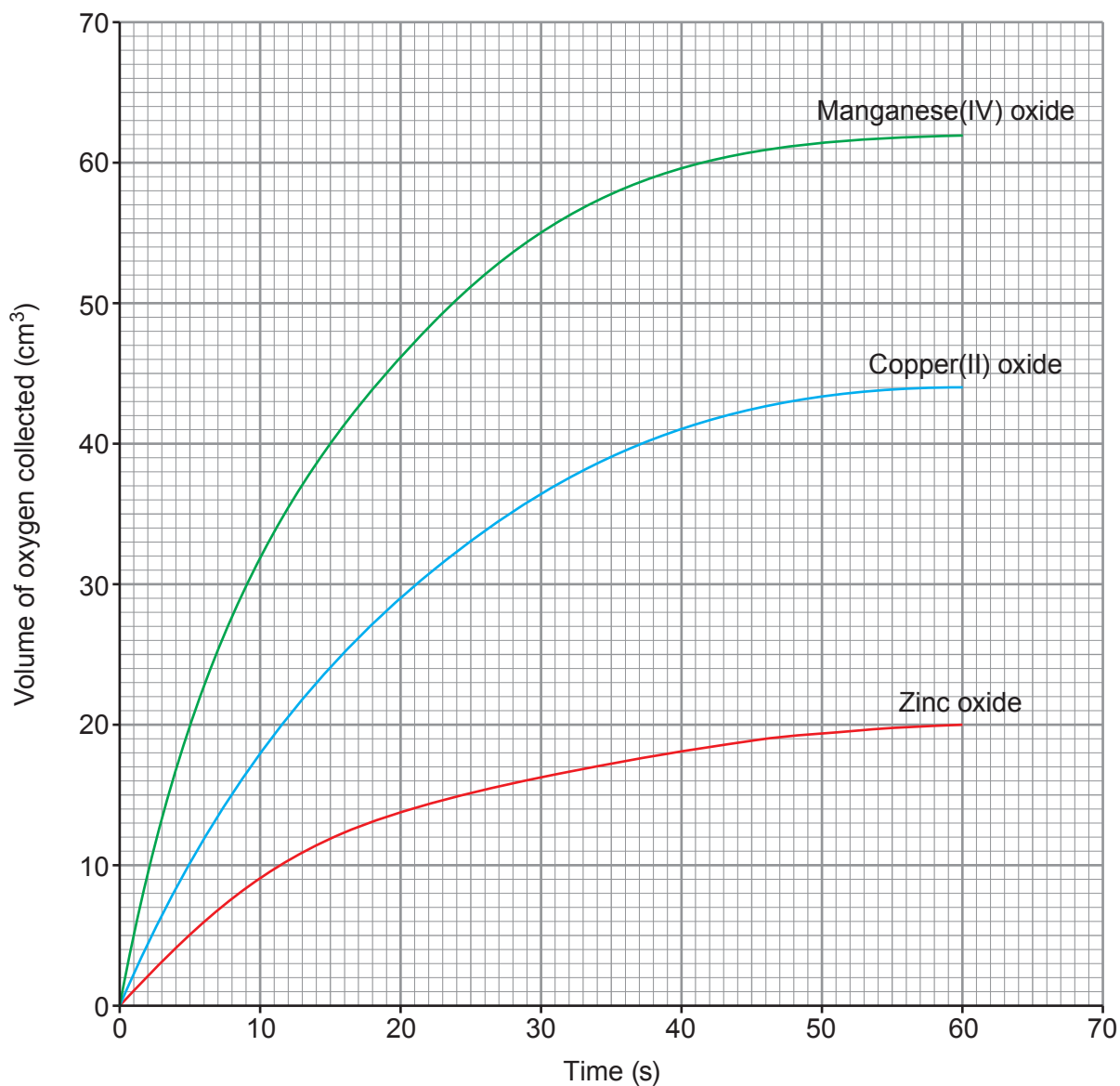


- (b) This reaction is very slow at room temperature.

A student suggests that the rate of reaction can be increased by adding a catalyst. State **one other** way that the rate of reaction could be increased. [1]

.....

- (c) Students investigated the decomposition of hydrogen peroxide using three different catalysts. The catalysts used were manganese(IV) oxide, copper(II) oxide and zinc oxide. Their results are shown on the graph below.



Use the graph to answer the following questions.

- (i) State the volume of oxygen collected in 10 seconds when manganese(IV) oxide is used. [1]

Volume = cm³

- (ii) State the time taken to collect 20 cm³ of oxygen when copper(II) oxide is used. [1]

Time = s

- (iii) One student says that zinc oxide is the best catalyst to use.
Explain whether the student is correct.

[2]

.....

.....

.....

- (d) The mass of catalyst used at the start of each experiment was 1.2 g.
State the mass of catalyst at the end of each experiment.

[1]

Mass = g

- (e) Milk used in cheese making is stored in the refrigerator at 4°C. Room temperature in the kitchen is 24°C.

- (i) Calculate the difference in temperature between the refrigerator and the kitchen.

[1]

Temperature difference = °C

- (ii) It is claimed that for every temperature decrease of 10°C, the milk will take twice as long to go sour. The milk sours in 10 hours when left out of the refrigerator in the kitchen. Calculate the time it takes the milk to sour in the refrigerator.

[2]

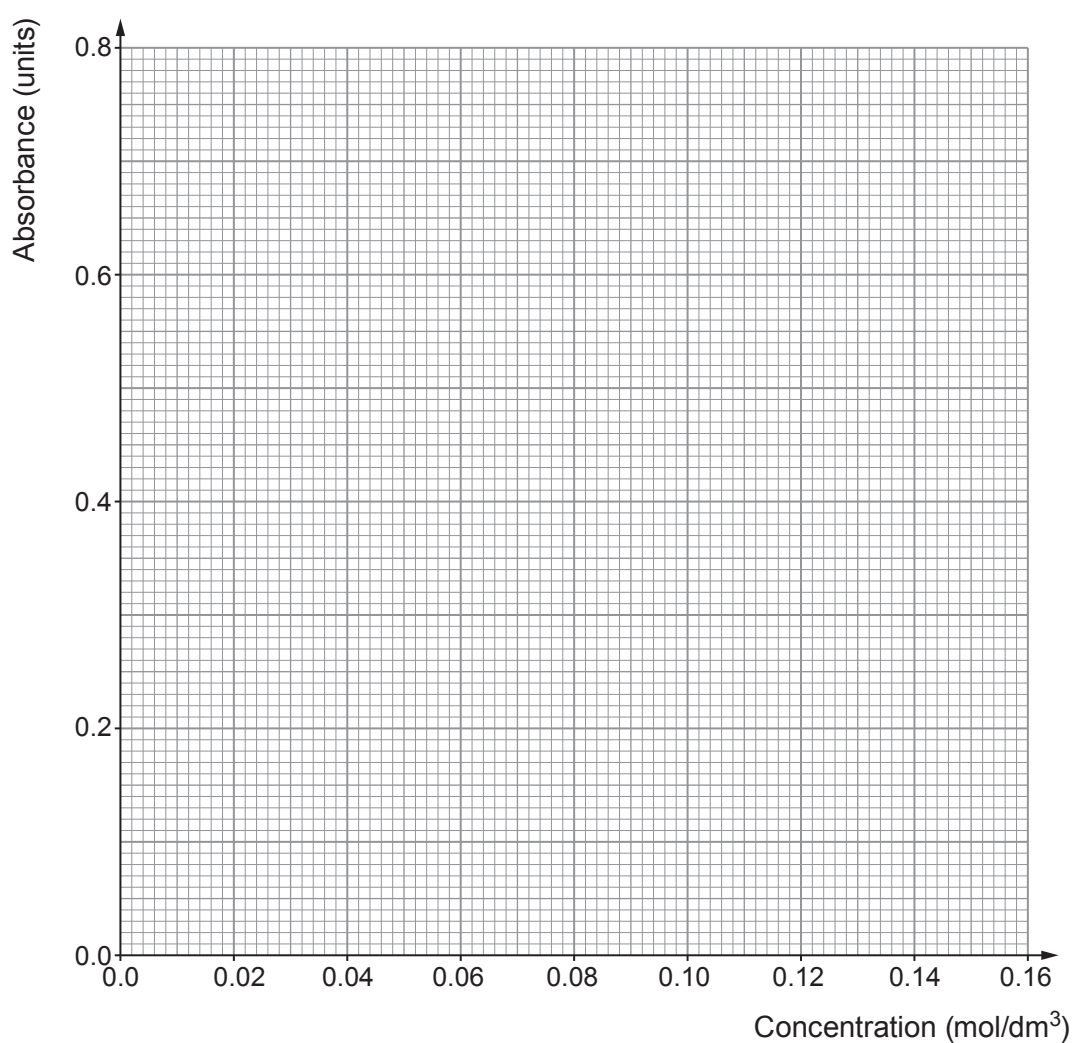
Time = hours

3. An experiment was carried out to find the concentration of beta-carotene in a solution using colorimetry. The results are shown in the table below.

Concentration (mol/dm^3)	Absorbance (units)
0.02	0.1
0.04	0.2
0.08	0.4
0.12	0.6
0.16	0.8

- (a) Use the data to plot a graph on the grid below and draw a suitable line.

[3]



- (b) Describe how the absorbance of beta-carotene changes with concentration. [1]

.....

- (c) Use your line to predict the concentration of beta-carotene with an absorbance of 0.3 units. [1]

Concentration = mol/dm³

- (d) It is claimed that if the concentration is doubled then the absorbance always doubles. Explain whether you agree with this statement using data from the table. [2]

.....

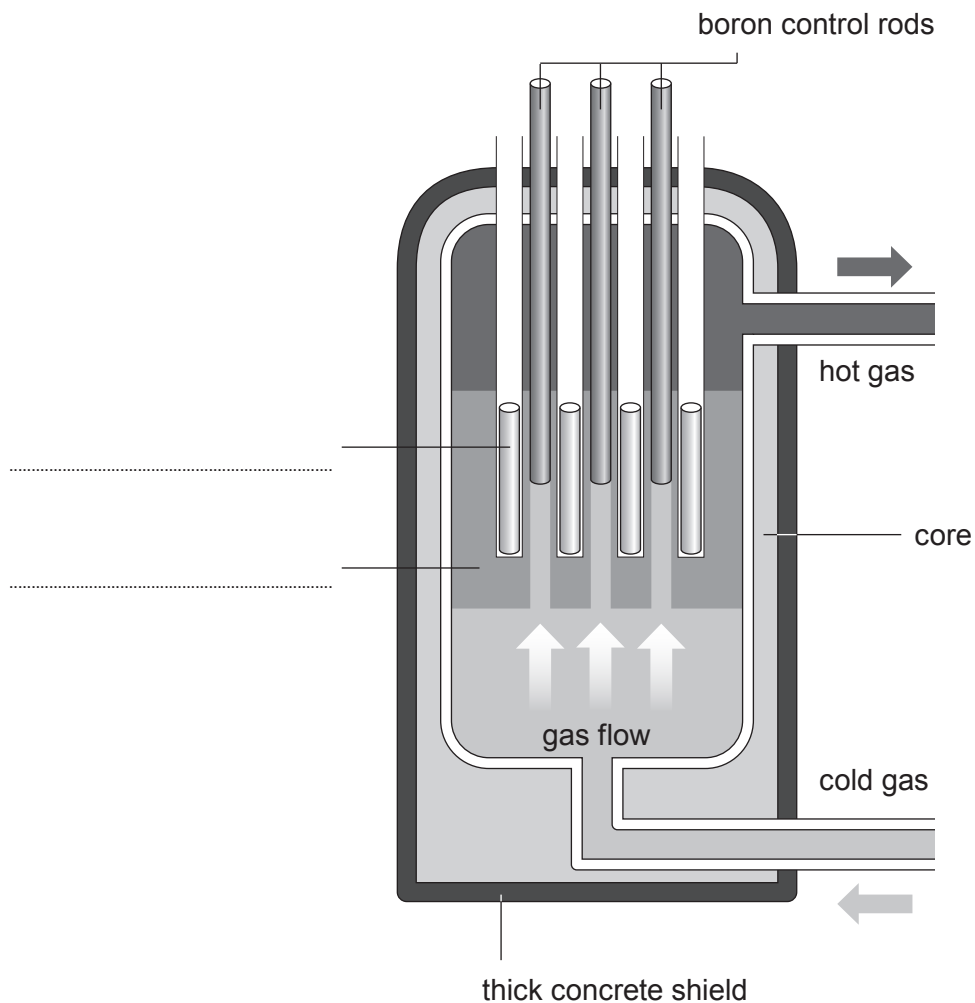
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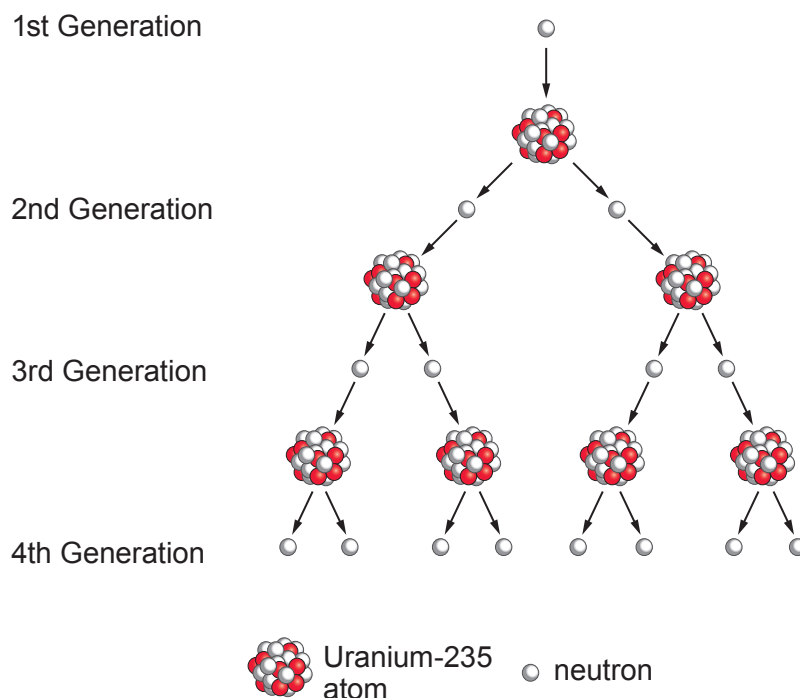
4. Nuclear fission occurs when a nucleus of uranium is split after absorbing a neutron. It is the process used in a nuclear reactor like the one shown in the diagram below.



(a) Complete the labelling of the diagram.

[2]

- (b) The diagram below shows an uncontrolled chain reaction of the decay of uranium.



- (i) Calculate the number of 5th generation neutrons. [1]
- (ii) Explain how the number of neutrons in your answer above would change if the control rods are lowered further into the reactor. [2]

.....

.....

.....

- (iii) State the danger of an uncontrolled chain reaction in a nuclear reactor. [1]

.....

- [6 QÉR]

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6

6. In 2016, the United Kingdom had a total area of 508 thousand hectares of land which was farmed organically.

- (a) Tick (✓) the **three** boxes next to the correct statements about the advantages of organic farming. [3]

The environment benefits because natural habitats are less threatened.

☐

Production of food is cheaper.

☐

Organic crops grow quicker.

☐

Artificial fertilisers are not used.

☐

Less labour involved in collection of organic food.

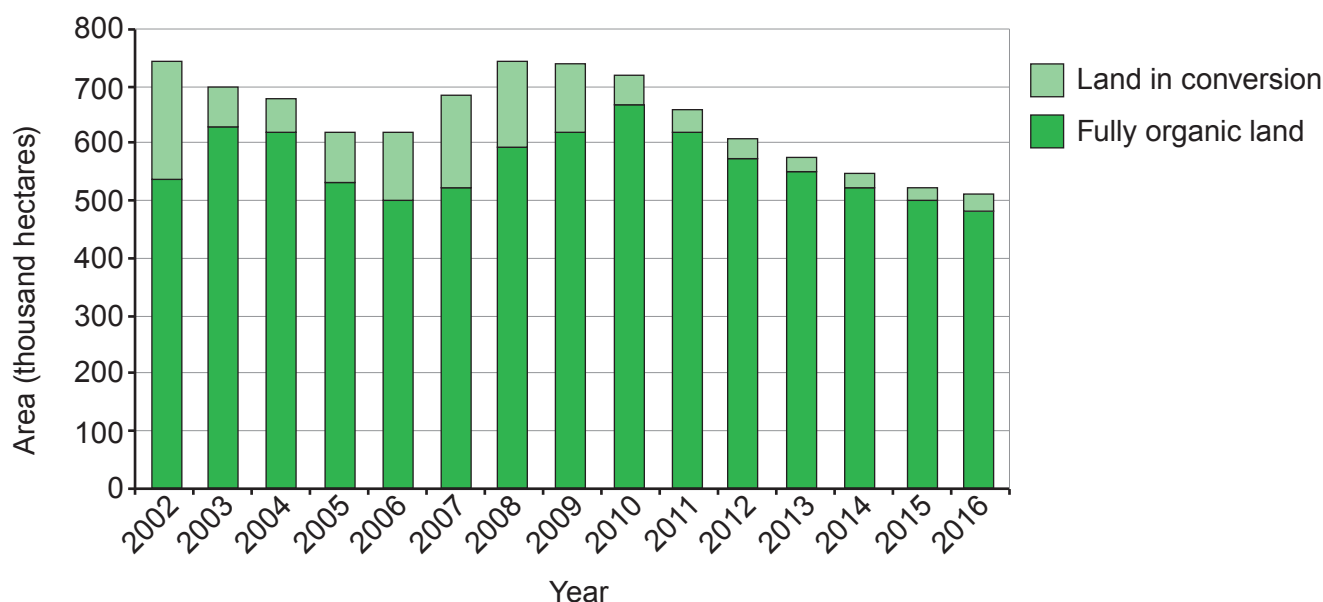
☐

Less harm to bees and other insects as less chemicals are used.

☐

- (b) Since 2005, farmers who use intensive farming methods have been encouraged to change to organic farming. Before an area can be considered as fully organic, it must undergo a conversion process.

The chart below shows how the area of fully organic land and land in conversion has changed since 2002.



Use the data in the chart to state whether there has been an increase in the area of land in conversion since 2005. [2]

.....

.....

.....

- (c) The table below shows the area of fully organic land used to produce crops from 2013 to 2016.

	Area of fully organic land use, 2013 to 2016 (thousand hectares)			
	2013	2014	2015	2016
Cereals	43.7	42.2	39.6	38.4
Other crops	7.6	7.3	6.9	7.3
Fruit & nuts	2.1	2.1	1.9	1.9
Vegetables	11.3	9.4	10.4	10.2
Total	64.7	61.0	58.8	

Calculate the change in area of fully organic land used to produce these crops from 2013 to 2016. [2]

Change in area = thousand hectares

- (d) The table below shows how organic livestock numbers have changed from 2013 to 2016.

	Organic livestock numbers 2013 to 2016 (thousand)			
	2013	2014	2015	2016
Cattle	283.3	304.1	291.5	296.4
Sheep	999.2	954.9	844.6	840.8
Pigs	30.2	28.3	30.0	31.5
Poultry	2487.6	2398.7	2560.2	2821.2

- (i) State **one** ethical argument against the intensive farming of livestock. [1]

.....

- (ii) Explain whether the data in the table supports that this ethical argument has changed livestock farming practices. [2]

.....

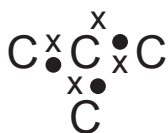
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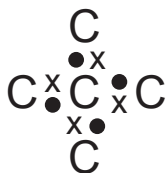
7. Carbon exists in many forms. Each carbon atom has 4 electrons available for bonding.

(a) Four different dot and cross diagrams are shown below.

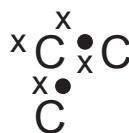
A



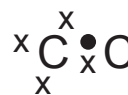
B



C



D



(i) State which diagram shows the bonding found in diamond. [1]

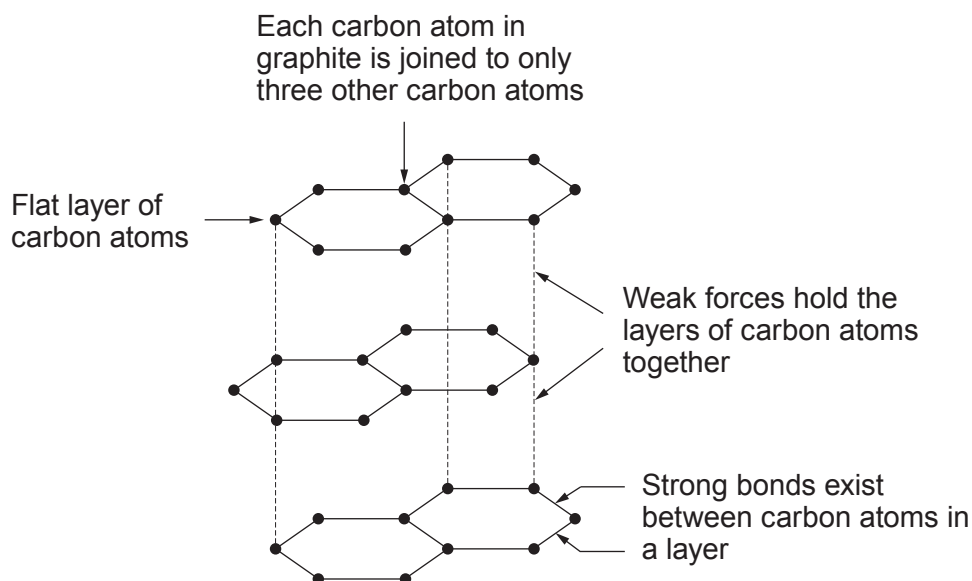
(ii) Give a reason for your choice. [1]

.....

(iii) Name the type of bond found between carbon atoms in diamond. [1]

.....

- (b) Carbon also exists in the form of graphite. The diagram shows the giant molecular structure of graphite.



- (i) Explain why graphite conducts electricity.

[2]

.....

.....

.....

- (ii) Explain why graphite is slippery.

[2]

.....

.....

.....

8. Photosynthesis is the process by which green plants produce their own glucose.

(a) Complete the word equation for photosynthesis below.

[1]

..... + → glucose + oxygen

(b) State **three** different ways green plants use the glucose they produce.

[3]

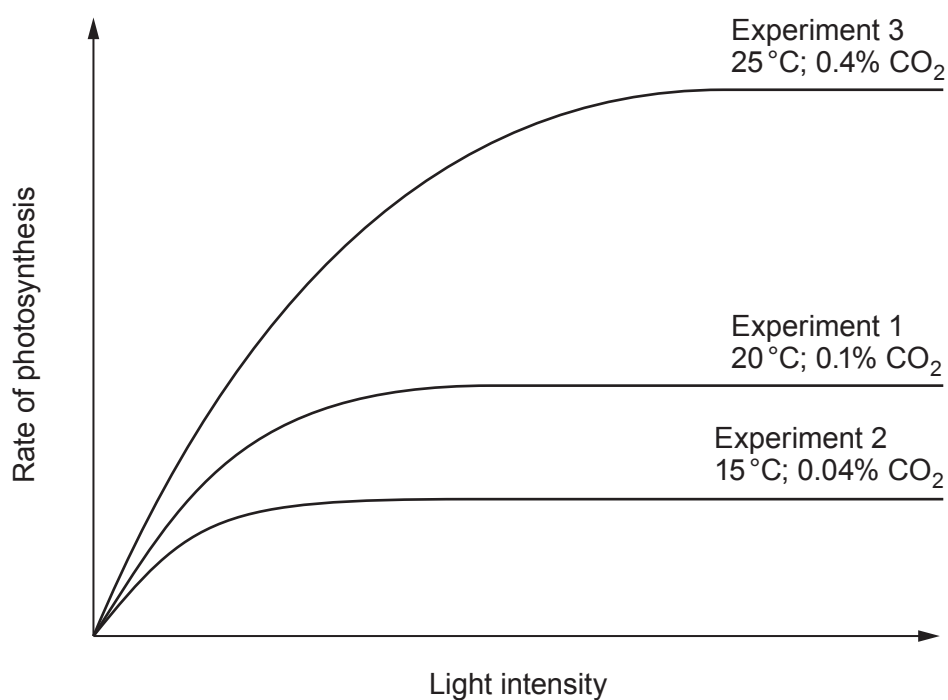
.....

.....

.....

.....

(c) The rate of photosynthesis is affected by light intensity, temperature and carbon dioxide concentration as shown in the graph below.



Use the information in the graph to describe how differences in light intensity, temperature and carbon dioxide concentration affect the rate of photosynthesis.

[3]

.....

.....

.....

.....

.....

9. (a) Food poisoning is a notifiable disease in Wales. Doctors are required to report every case of food poisoning.

(i) Explain how some bacteria cause food poisoning. [2]

.....

.....

.....

(ii) State **two** symptoms of food poisoning. [1]

.....

.....

(iii) State **two** different types of precaution that can be taken when **preparing** food to reduce the risk of food poisoning. [2]

.....

.....

.....

- (b) **Figure 1** shows the number and rate of reported cases of food poisoning in Wales between 1992 and 2016.

Figure 1

Year	Total number of reported cases	Rate (number per 100 000 population)
1992	3 590	124.5
1998	5 946	205.0
2000	4 716	162.2
2006	4 301	145.4
2010	4 980	165.6
2014	4 516	146.1
2016	4 504	145.7

The percentage of reported cases of food poisoning during each quarter of a year is shown in **Figure 2**.

Figure 2

Quarter	% of total for year		
	2012	2014	2016
Jan-Mar	18	17	19
Apr-Jun	26	26	25
Jul-Sep	32	32	33
Oct-Dec	24	25	23

Figure 3 shows the rate of food poisoning amongst females by age group in 2016.

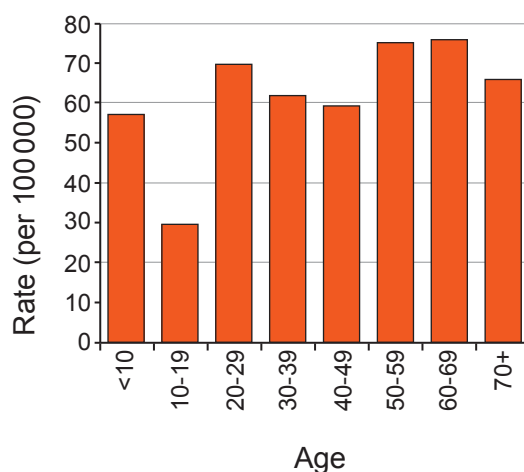
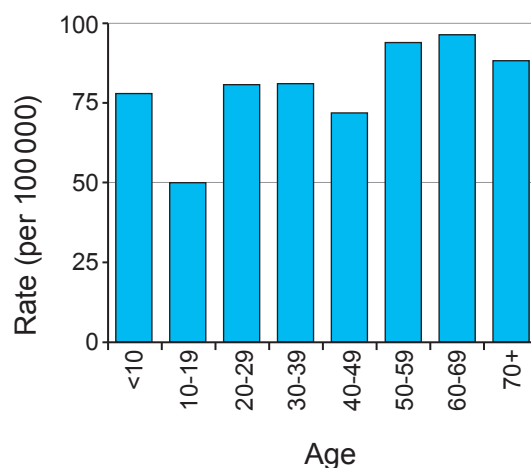
Figure 3

Figure 4 shows the rate of food poisoning amongst males by age group in 2016.

Figure 4



Use the data in **Figures 1 to 4** to answer the following questions.

- (i) Since 1998, egg-laying hens have been vaccinated against a species of *Salmonella*. Explain how this has affected the number of reported cases of food poisoning. [2]

.....

.....

.....

- (ii) Explain the variation in the percentage of reported cases of food poisoning in each of the quarters in a year. [3]

.....

.....

.....

.....

.....

- (iii) It is suggested that the rate of food poisoning does not depend on gender or age. Explain whether this suggestion is confirmed by the data. [2]

.....

.....

.....

END OF PAPER

THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1												4 He Helium 2							
7 Li Lithium 3		9 Be Beryllium 4		11 B Boron 5 12 C Carbon 6 13 Al Aluminium 14 Si Silicon 15 P Phosphorus 16 S Sulfur 17 Cl Chlorine 18 Ar Argon								19 F Fluorine 9 20 Ne Neon 10							
23 Na Sodium 11		24 Mg Magnesium 12		27 Al Aluminium 28 Si Silicon 31 P Phosphorus 32 S Sulfur 35.5 Cl Chlorine 40 Ar Argon								35.5 Cl Chlorine 17 36 Kr Krypton 36							
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89		Key													

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3445U30-1



Z22-3445U30-1-R1

TUESDAY, 17 MAY 2022 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 3: Food, Materials and Processes

FOUNDATION TIER

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	10	
3.	8	
4.	6	
5.	9	
6.	14	
7.	19	
Total	75	

3445U301
01

ADDITIONAL MATERIALS

A calculator.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **4** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of the examination paper.



JUN223445U30101

Answer **all** questions in the spaces provided.

1. The production and processing of dairy products requires a number of processes.

- (a) Statements **A-E** below show the main stages of making yoghurt. However the stages are **not** in the correct order.

A

The equipment is sterilised

B

Samples are taken to see if the yoghurt is ready

C

The milk is pasteurised by heating

D

The bacterial culture is added to the pasteurised milk and left at 46 °C for about four hours

E

Flavours and colours are added, then the yoghurt is packaged

- (i) Use the letters **B, C, D**, to arrange the stages in the correct order in the boxes below. [2]

A				E
----------	--	--	--	----------

- (ii) Underline the type of microorganism used in this process. [1]

fungus bacterium virus

- (iii) State **one** reason why the equipment should be sterilised before use. [1]

.....

- (iv) State why the milk is pasteurised during the process. [1]

.....



- (b) (i) Whole milk, semi-skimmed milk and skimmed milk are available in supermarkets. Complete the following sentence. [2]

Semi-skimmed milk is made by skimming from
..... milk.

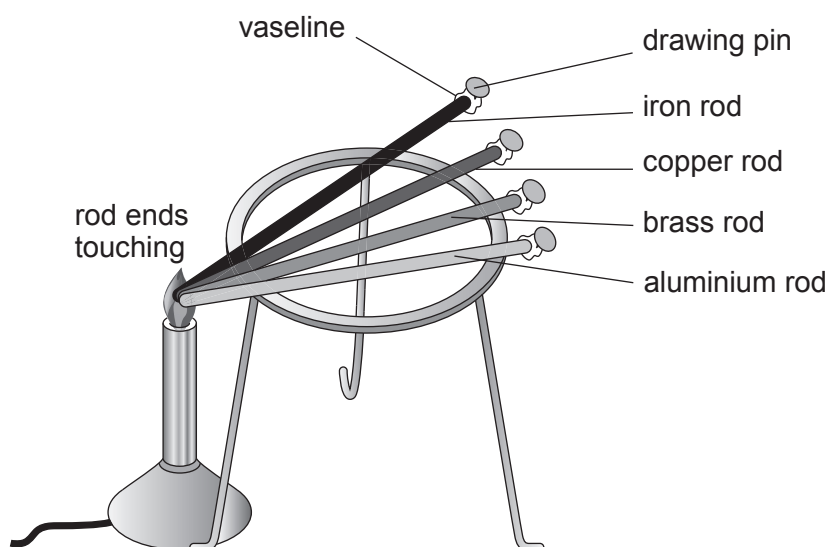
- (ii) Complete the following sentences about homogenisation of milk by underlining the correct word in the brackets.

The milk is pumped at high (**pressure** / **temperature** / **volume**) through very narrow tubes breaking up the fat globules. This prevents a layer of (**butter** / **cheese** / **cream**) forming on the milk. [2]



2. Aluminium, copper and iron are metals. Brass is a metal alloy. These materials have many uses.

The thermal conductivity of metals can be investigated using the apparatus below. The time taken for a drawing pin to fall from each metal rod is measured.



- (a) (i) State the independent variable in this experiment. [1]

.....

- (ii) State **two** variables that are controlled in this experiment. [2]

.....

.....

- (iii) Describe how the thermal conductivity is related to the dependent variable. [2]

.....

.....



- (b) Two students obtained the results shown in the table below.

Metal	Time taken for pin to drop (s)
iron	17
copper	7
brass	12
aluminium	5

- (i) Use the results shown in the table above to place iron, copper, brass and aluminium in order of conductivity. [1]

Best conductor



Poorest conductor

- (ii) The following table shows the theoretical values of the thermal conductivity of the metals.

Metal	Thermal conductivity (units)
iron	80.4
copper	401
brass	109
aluminium	210

Use the information above to explain whether the theoretical data agrees with experimental data. [3]

Best conductor



Poorest conductor

.....

.....

.....

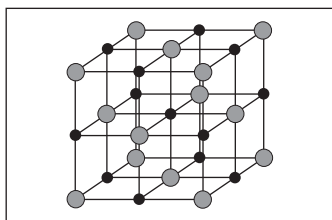
- (c) Mercury is a metal which has a melting point of -39°C . Suggest why the students could not use mercury in this investigation. [1]

.....

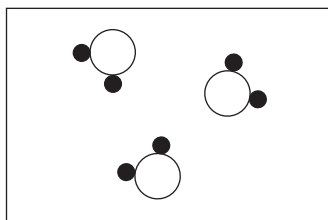
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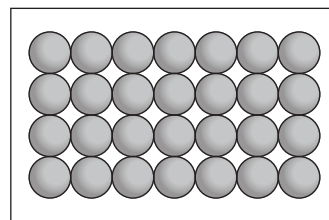
3. The following diagrams show the structure of substances.



A (solid)



B (gas)



C (solid)

- (a) State which of the structures (**A**, **B** or **C**) is an ionic compound. Give **one** reason for your answer. [2]

Letter:

Reason:

.....
.....

- (b) Sodium chloride is an ionic compound which consists of sodium and chloride ions.

Explain why sodium chloride will not conduct electricity when it is a solid but will conduct electricity when it is dissolved in water. [2]

.....
.....



- (c) Some of the materials used in sporting equipment are shown in the table below.

Material	Thermal conductivity (units)	Melting point (°C)	Density (g/cm ³)	Tensile strength (MPa)
carbon fibre	21	3650	1.78	4137
aluminium	210	660	2.71	276
lead	35	327	11.3	12

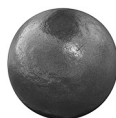
Use the data in the table to answer the following questions.

- (i) Carbon fibre is an example of a composite material used to make bicycle frames for use in cycle races.

State **two** reasons why modern bicycle frames are made of carbon fibre rather than aluminium. [2]

1.
2.

- (ii) A shot put contains lead.



Use the equation

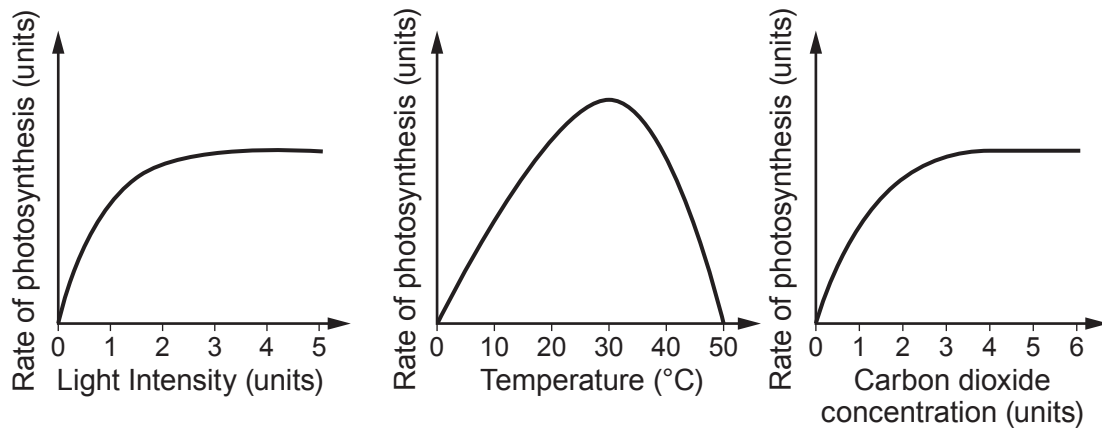
$$\text{mass} = \text{density} \times \text{volume}$$

to find the mass of lead in a shot put that has a volume of 700 cm³. [2]

mass = g

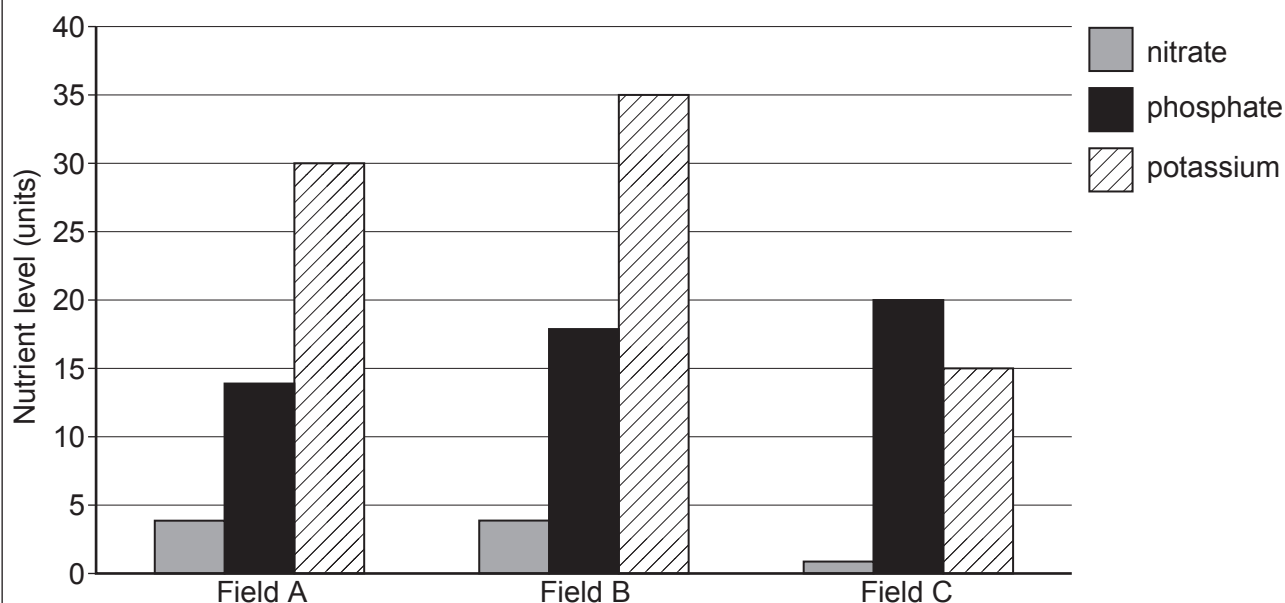


- Their results are shown in the graphs below.



Use the graphs to describe how the rate of photosynthesis is affected by each of these factors. [6 QER]

5. A farmer wanted to grow soybean crops on three fields. She analysed soil samples for nitrate, phosphate and potassium. The results are shown below.



Soybeans will show healthy growth if the nutrient levels in the soil are above a threshold value. The plants will show signs of mineral deficiency if the nutrient levels in the soil are below these values.

The nitrate, phosphate and potassium threshold values for soybeans are shown in the table below.

Nutrient	Threshold value (units)
nitrate	2
phosphate	16
potassium	28

- (a) Use the information above to answer the following questions.

- (i) State which field would be the most suitable to grow soybean crops. Give a reason for your answer. [2]

Field:

Reason:

.....



- (ii) State which field would result in soybeans with yellow leaves and stunted growth. Give a reason for your answer. [2]

Field:

Reason:

- (iii) State how the farmer ensures that her soybean plants do not suffer from mineral deficiencies in any field used to grow them. [1]

- (b) Some countries have banned the growing of genetically modified (GM) crops. The table below gives some arguments for and against the use of GM crops. Place a tick (✓) next to **two** arguments **against** growing GM crops. [2]

Argument	Against (✓)
cheaper food	
increased crop yield	
could cross-pollinate with weeds	
long-term effects unknown	

- (c) Selective breeding is used by large companies to increase the yield of soybeans. State **two** disadvantages of using crops developed by a selective breeding process. [2]

1.

2.



6. Despite safety precautions, industrial accidents occur.

- (a) Methyl isocyanate (MIC) was involved in the Bhopal disaster that led to the deaths of thousands of people. A highly toxic cloud of MIC was released from chimneys at the chemical plant. On the night of the disaster it reacted with water that leaked into a storage tank, causing a valve to blow.

The structural formula of MIC is $\text{H}_3\text{C}-\text{N}=\text{C}=\text{O}$.

- (i) Complete the table below to calculate the total energy required to break all the bonds in 1 mole of the reactants, MIC and water. [5]

$\text{H}_3\text{C}-\text{N}=\text{C}=\text{O}$			
Bond type	Number of bonds	Energy to break one bond (kJ/mol)	Energy to break all of each bond type (kJ/mol)
H—C	3	413	
C—N	1	305	305
N=C	1	615	
C=O	1		745

H_2O			
Bond type	Number of bonds	Energy to break one bond (kJ/mol)	Energy to break all of each bond type (kJ/mol)
H—O	2	467	

total energy to break **all** bonds = kJ/mol

- (ii) A similar calculation shows the energy released when bonds are formed in the products is greater than your answer above.

Complete the paragraph below by underlining the correct word(s) in the brackets. [3]

When a chemical reaction happens, energy is transferred to or from the surroundings. Energy is transferred to the surroundings if the energy released is **(less than / equal to / greater than)** the energy absorbed. This is called **(an endothermic / a combustion / an exothermic)** reaction. The temperature of the surroundings **(decreases / stays the same / increases)**.



- (iii) The Bhopal disaster was caused because of thermal runaway.

Explain what is meant by the term 'thermal runaway'.

[2]

.....

.....

.....

- (b) Spinach grown in north-eastern Japan was contaminated by radioactivity from the Fukushima nuclear power station accident in March 2011. Radioactive dust deposited iodine-131 atoms on spinach leaves. Its activity was 8200 units.

No spinach was allowed to be used for 96 days.

Iodine-131 has a half-life of 8 days.

- (i) The initial activity of the iodine-131 was 8200 units. Calculate the activity after 8 days.

[1]

activity units

- (ii) Calculate the number of half-lives in 96 days.

[1]

number =

- (iii) Calculate the activity after 96 days.

[2]

activity units



7. Criminals will try to hide their activities but forensic scientists, using biological and chemical testing, are able to find evidence that reveals what happened.

(a) DNA profiling is commonly used to identify criminals.

(i) State where DNA is found in the cell.

[1]

.....

(ii) There were three suspects in a crime. Suspect 1 did not have an alibi but suspects 2 and 3 did. Initially it was thought that suspect 1 was the criminal. DNA samples were collected from the crime scene.



Compare the DNA samples above to explain whether having an alibi or not is sufficient to decide on guilt.

[2]

.....

.....

.....

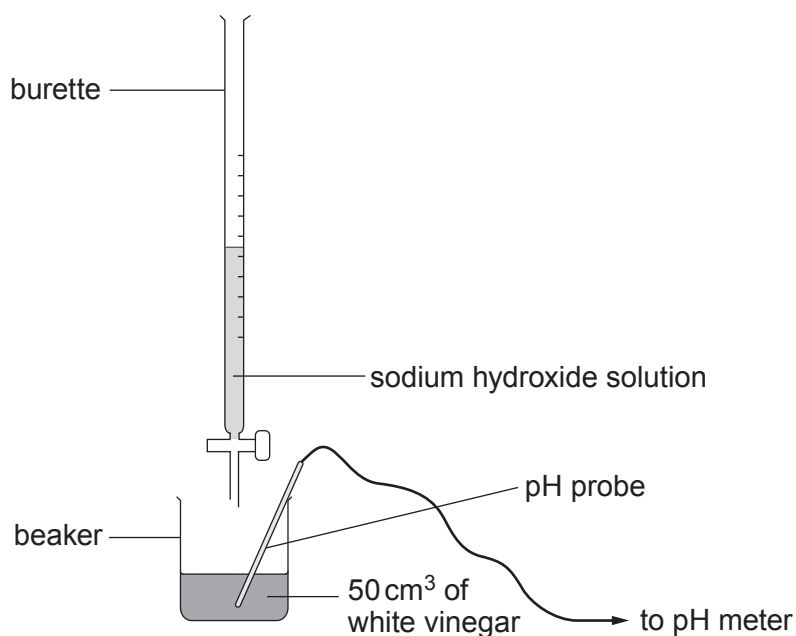
(iii) State **one other** use of DNA profiling.

[1]

.....



- (b) It is suspected that supplies of vinegar have been tampered with by adding a strong acid. White vinegar usually contains ethanoic acid at a concentration of 0.01 mol/dm^3 . Samples of the vinegar are taken and tested by titrating against 0.5 mol/dm^3 sodium hydroxide solution using the apparatus below.



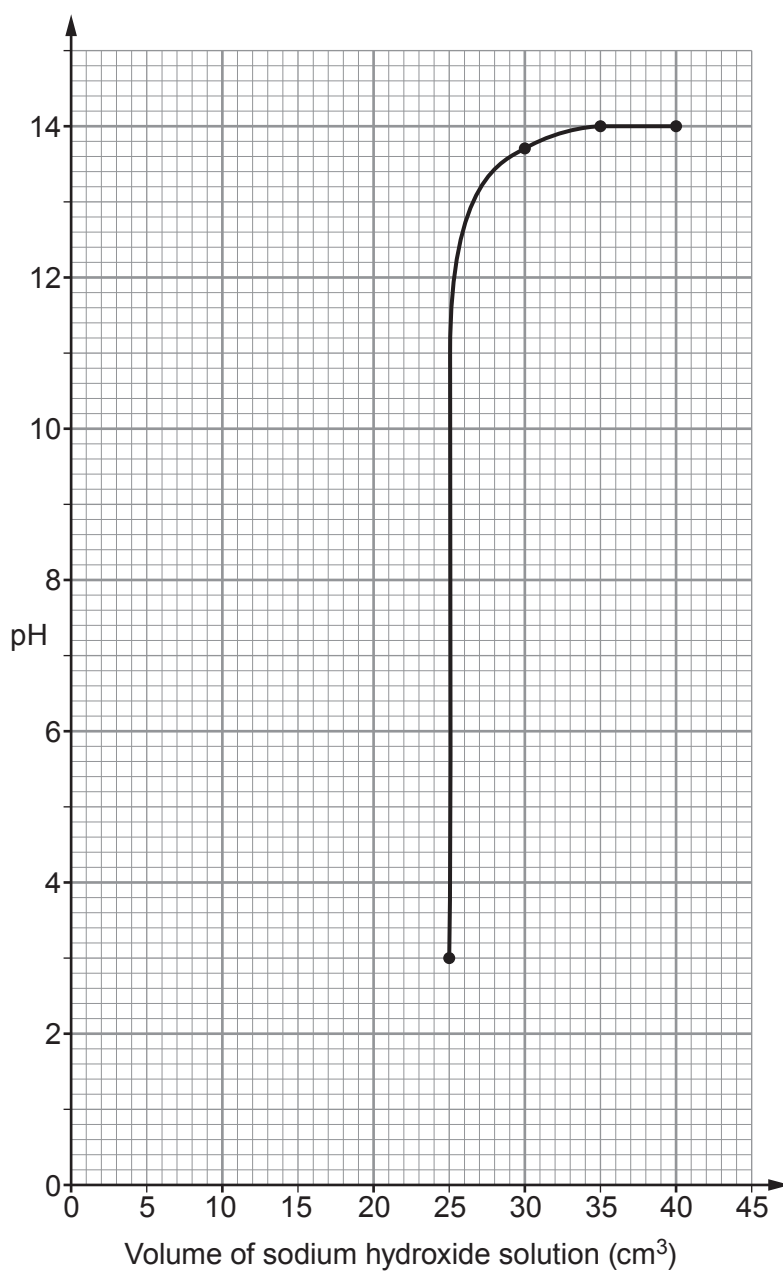
A pH meter is used to monitor the pH of the solution in the flask. The results are shown in the table below.

Volume of sodium hydroxide solution (cm^3)	pH reading
0	1.5
5	1.6
10	1.7
15	1.9
20	2.2
25	3.0
30	13.7
35	14.0
40	14.0



- (i) Use the data in the table to complete the graph on the grid below.

[3]



- (ii) Use the graph to find the volume of sodium hydroxide solution required to neutralise the white vinegar.

[1]

volume = cm³

- (iii) Use the information on pages 16 and 17 and the equation

$$\text{concentration of white vinegar} = \frac{\text{concentration of sodium hydroxide} \times \text{volume of sodium hydroxide}}{\text{volume of white vinegar}}$$

to determine whether or not the samples had been tampered with. [3]

-
.....
- (iv) Complete the word equation for the reaction between an acid and a base. [2]

acid + base →

- (v) Explain why taking multiple readings will increase the accuracy of the result. [2]

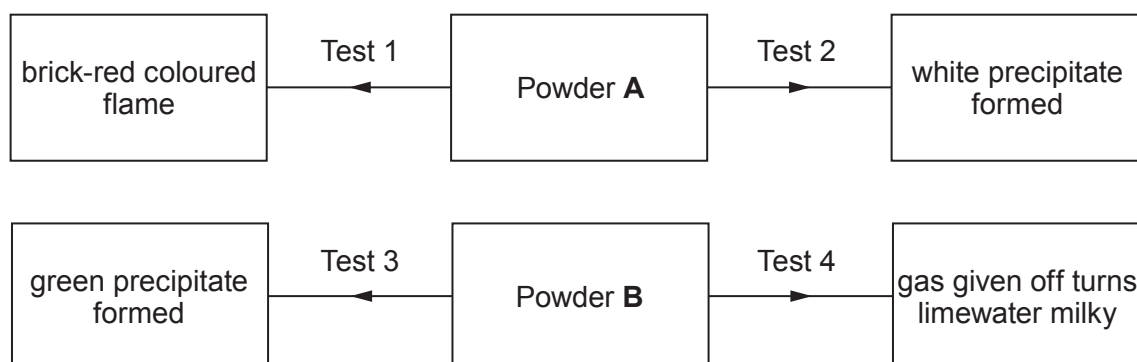
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- (c) Powders **A** and **B** are collected at a crime scene. The following tests are used to identify them.

Test	Action
1	Flame test
2	Add silver nitrate solution
3	Add sodium hydroxide solution
4	Add dilute hydrochloric acid and bubble gas through limewater

The results are given below.



Identify powders **A** and **B**.

[4]

Powder **A**:

Powder **B**:

END OF PAPER





Group

2

3

4

5

9

2

0

20

Key

A_r	Symbol	Name	Z
-------	--------	------	---

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3445U30-1



S23-3445U30-1

TUESDAY, 16 MAY 2023 – MORNING

APPLIED SCIENCE (Double Award)

UNIT 3: Food, Materials and Processes

FOUNDATION TIER

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	11	
3.	12	
4.	10	
5.	13	
6.	5	
7.	14	
Total	75	

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator, a pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 4(a) is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer is incorrect.

The Periodic Table is printed on the back cover of the examination paper.



JUN233445U30101

Answer **all** questions in the spaces provided.

1. Ash dieback disease is a fungal infection that is common in ash trees in Wales.

The disease causes the leaves to blacken and wilt, so eventually the ash tree has no leaves.

The photographs show healthy ash leaves and those affected by ash dieback.



- (a) Photosynthesis occurs in green leaves.

- (i) Use word(s) from the box to complete the paragraph below. You may use each word once, more than once or not at all. [5]

stomata	nitrogen	chlorophyll	oxygen	glucose
	water	carbon dioxide		

Leaves use to absorb light energy.

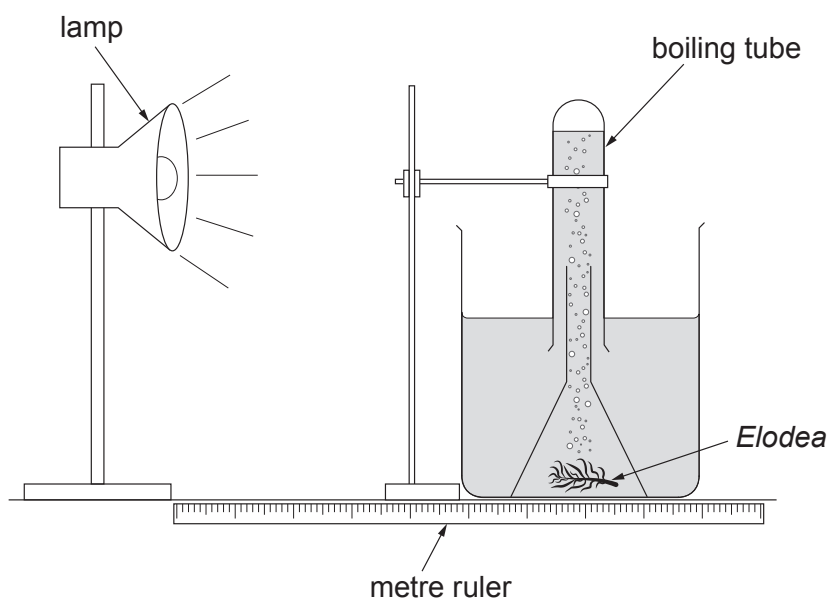
They convert and into
..... and

- (ii) Complete the following sentence by underlining the correct word in the brackets. [1]

Ash dieback disease affects the growth of an ash tree as the leaves are unable to make (**food** / **water** / **nitrogen**).



- (b) The following apparatus is used to investigate how light intensity affects the rate of photosynthesis of an *Elodea* plant.



The method used is described below.

1. Add **one spatula** of sodium hydrogen carbonate to a beaker containing 200 cm^3 of water at 20°C , to supply carbon dioxide.
2. Place the *Elodea* in the beaker.
3. Completely fill a boiling tube with water and carefully place over the end of the funnel with the end under the water, clamp into place.
4. Place the lamp 5 cm away from the *Elodea*.
5. Start the stopwatch and record the **number of bubbles** of oxygen produced in a **few seconds**.
6. Repeat the experiment with the lamp 10 cm, 15 cm, 20 cm, 25 cm and 30 cm from the apparatus.

- (i) Steps 1 and 5 could lead to uncertainties. In the table below, circle **one** improvement in each row that you would use.

[3]

Current method	Improvement	
add one spatula	add 1 kg	add 1 g
record the number of bubbles	collect the gas in a beaker	collect the gas in a measuring cylinder
produced in a few seconds	produced in 1 minute	produced in 1 hour

- (ii) State how you would change step 6 to investigate the effect of temperature on photosynthesis.

[1]

.....

.....

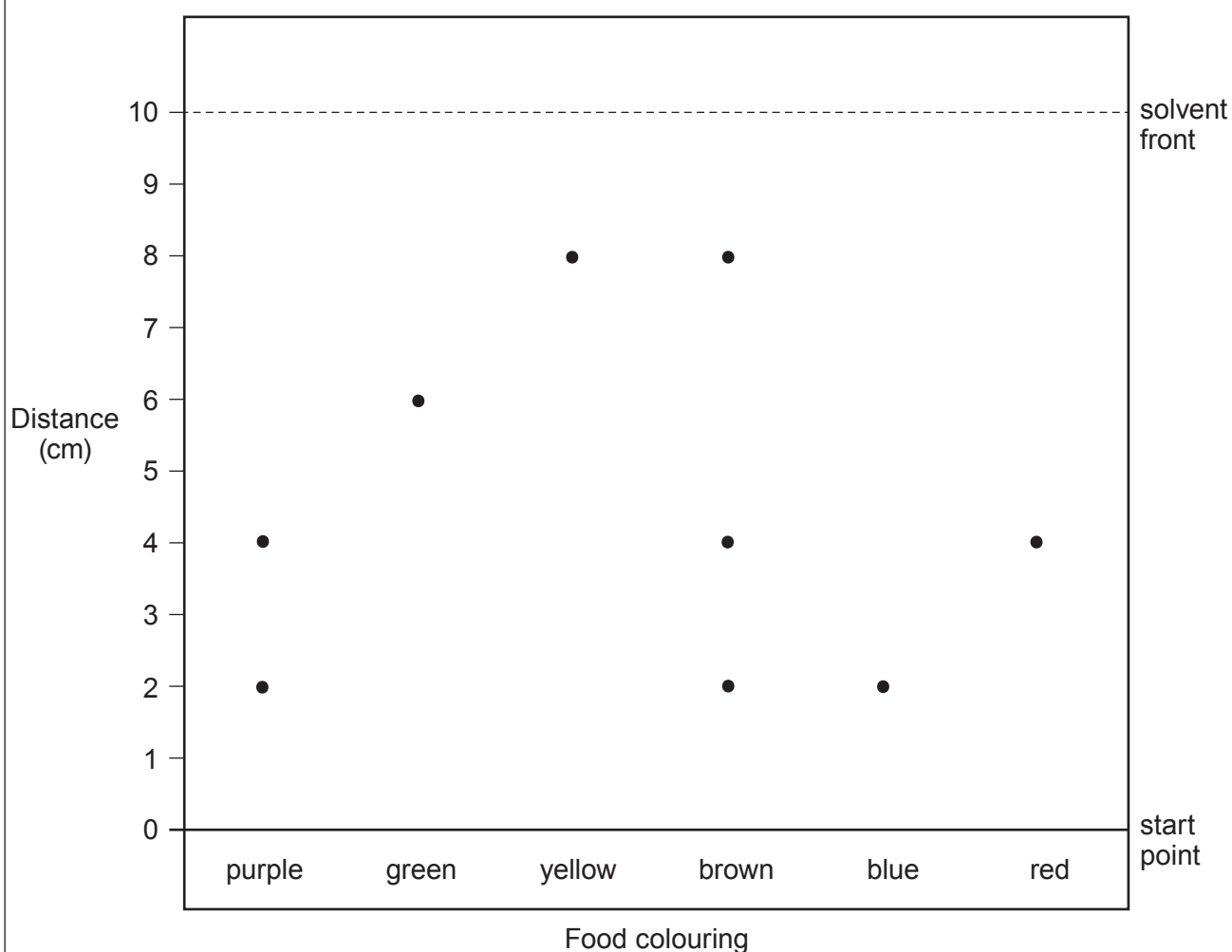
10



2. Analytical scientists work in a wide range of different industries and agencies.

- (a) The Food Standards Agency (FSA) funded a project to analyse the colours used in food and drinks.

The diagram below shows a chromatogram of the dyes present in different food colourings.



- (i) It was thought that the dye in the blue colouring would not be present in other colourings.
Explain whether you agree.

[2]

.....

.....



(ii) Use the equation:

$$R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by solvent}}$$

to calculate the R_f value for the dye in the **red** colouring.

[3]

$R_f =$

- (b) Analytical scientists also identify unknown compounds.
There were three bottles, each containing a solution of one of the following compounds.

copper(II) sulfate

lithium carbonate

sodium chloride

- (i) State which of these compounds would produce a yellow flame in a flame test. [1]

- (ii) I. State which of these compounds would produce bubbles of gas when added to hydrochloric acid. [1]

- II. The gas turns limewater milky. Name this gas. [1]

- (iii) 1 cm³ of silver nitrate solution is added to three different test tubes, each containing a solution of one of the compounds in the box above.

- I. State which of these compounds will give a white precipitate. [1]

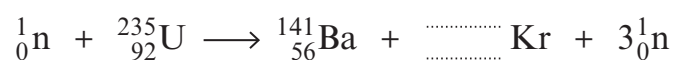
- II. Complete the risk assessment below for silver nitrate. [2]

Hazard	Risk	Control measure
silver nitrate is irritant	could cause eye damage if splashes in the eyes when	
		
		
		



3. Hunterston B is a nuclear power station in Scotland.

(a) The reaction in the reactor is shown in the equation below.



(i) **Complete** the equation. [2]

(ii) Circle the part of the reactor that slows down neutrons so they can be absorbed by uranium nuclei. [1]

moderator fuel rods concrete

(iii) State how many neutrons are produced by the fission of each uranium nucleus. [1]

.....

(b) Nuclear reactor 3 at the Hunterston B power station was shut down after cracks were found in the graphite bricks. This made it difficult for the control rods to move up and down.

(i) Explain the role of control rods. [2]

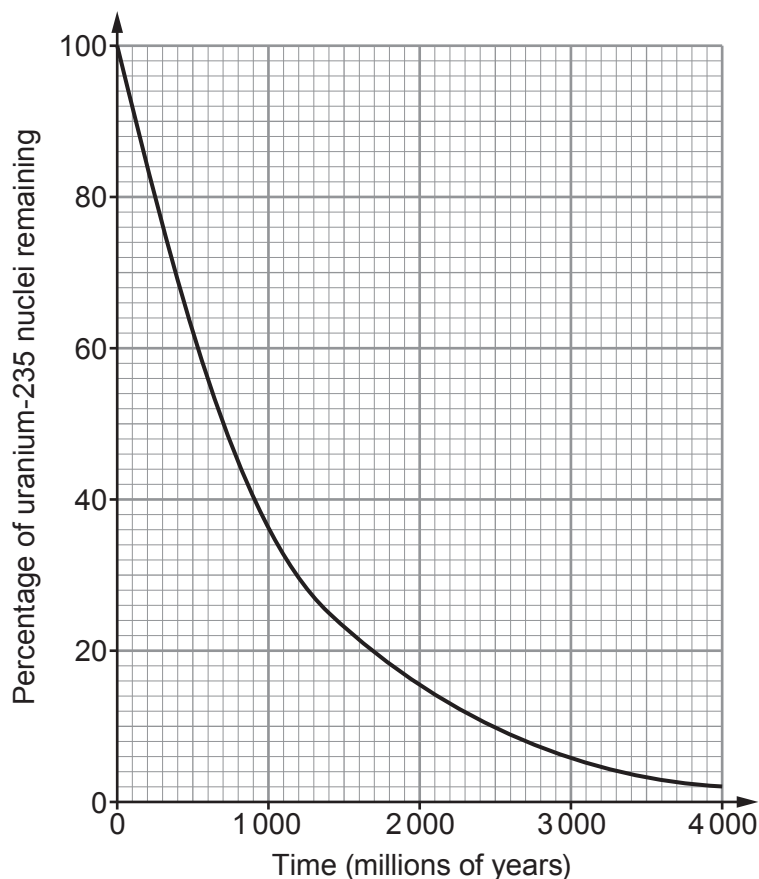
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.....
.....

(ii) An unusual event, such as an earthquake, might move the damaged graphite bricks so that the control rods are unable to move. Explain why this is dangerous. [2]

.....
.....
.....



- (c) After Hunterston B power station is decommissioned, the uranium-235 in the fuel rods will continue to decay at the rate shown in the graph below.



- (i) Use the graph to answer the following questions.

I. Find the time taken for the uranium-235 to decay from 100% to 50%. [1]

time = million years

II. State the half-life of uranium-235. [1]

half-life = million years

- (ii) The table shows the percentage of uranium-235 nuclei remaining after different half lives. Complete the table. [2]

Number of half-lives	0	1	2	3	4
Percentage of uranium-235 nuclei remaining	100		25		6.25



4. Heating is a common method used to preserve food.

- (a) **State three** methods, **other than heating**, in which the growth of bacteria can be slowed down or stopped in food products.

Explain how each method works.

[6 QER]

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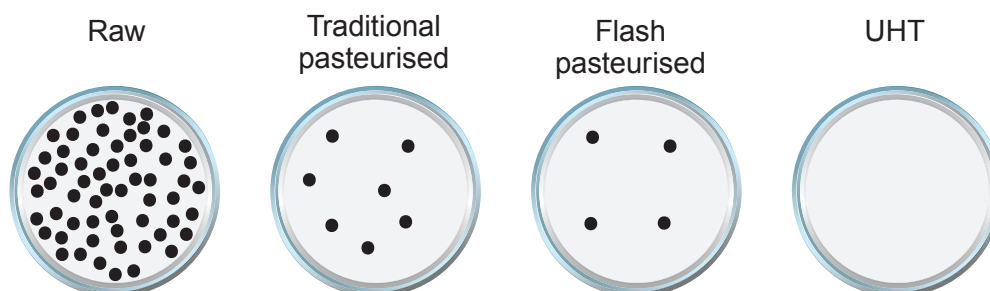
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- (b) Scientists investigated four types of milk. They monitored the growth of bacterial colonies on agar plates using 0.1 cm^3 from each type of milk over two days. Their results are shown below.



Type of milk	Heat treatment	Number of bacterial colonies from 0.1 cm^3 of milk	Number of bacterial colonies in a 40 cm^3 serving
raw	none	60	24 000
traditional pasteurised	63°C for 30 minutes	7	
flash pasteurised	78°C for 35 seconds		1 600
ultra-high temperature pasteurised (UHT)	135°C for 2 seconds	none	none

- (i) Complete the table.

[2]

Space for working

- (ii) It is claimed that unopened UHT milk can be stored on supermarket shelves at room temperature for six months. Explain whether you agree with the claim. [2]

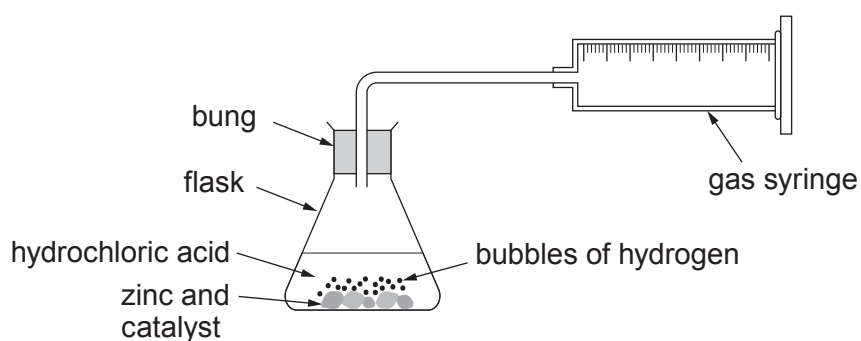
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5. The diagram below shows apparatus that can be used to investigate the rate of the reaction between large zinc pieces and hydrochloric acid. A catalyst was also added to the flask.

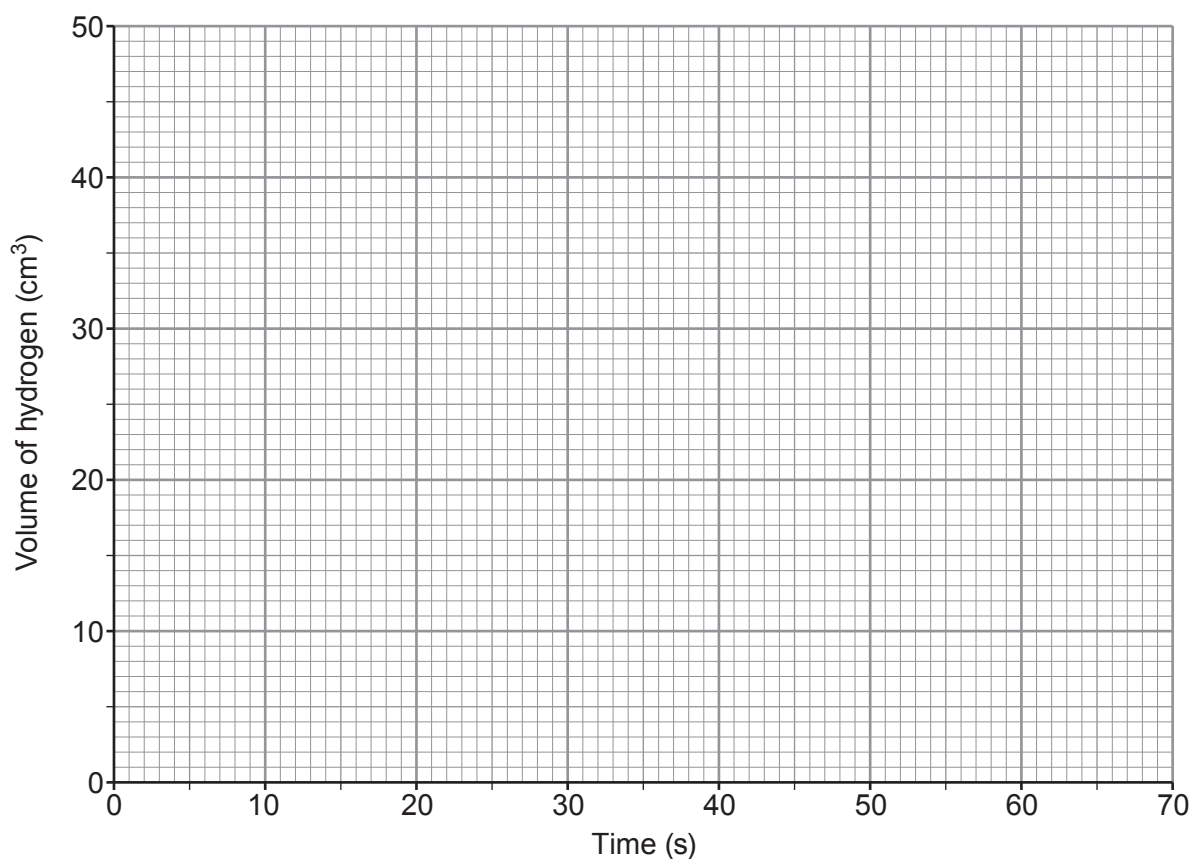


Large pieces of zinc were placed in excess dilute hydrochloric acid and the volume of hydrogen produced was recorded every 15 seconds. The results obtained are shown below.

Time (s)	0	15	30	45	60
Volume of hydrogen (cm ³)	0	28	40	48	48

- (a) (i) Plot the results on the grid below and draw a suitable curve.

[3]



(ii) Use your graph to answer the following questions.

I. Describe the relationship between volume of hydrogen produced and time.

[2]

.....

.....

.....

II. State the volume of hydrogen collected at 25 s.

[1]

volume = cm³

III. At what time did the reaction stop?

[1]

time = s

(b) (i) State the purpose of the catalyst.

[1]

.....

(ii) The mass of catalyst added at the start of the experiment was 1.5 g.
State the mass of catalyst remaining after 60 s.

[1]

mass = g

(iii) State how the **volumes in the table** would be different if the catalyst had not been added.

[1]

.....

(c) (i) State how the **volumes in the table** would be affected by using the same mass of zinc **powder** instead of large pieces of zinc.

[1]

.....

(ii) Explain how the **volumes in the table** would be affected if the original experiment was repeated at a higher temperature.

[2]

.....

.....

.....



6. (a) State, in terms of electrons, the difference between ionic and covalent bonding. [2]

.....

.....

.....

- (b) When magnesium reacts with oxygen, the ionic compound magnesium oxide is formed.

- (i) The outer electrons of magnesium and oxygen are shown below. Draw arrows to show the transfer of electrons when magnesium oxide is formed. [1]



- (ii) State the charges on the ions formed [2]

Mg O



7. Inos is a Welsh caravan manufacturer based in Denbighshire. The materials used to build modern caravans are very different to those used in early caravans.

Table 1 shows how the materials used in manufacturing the body of a caravan have changed over time.

Year	Materials used
1920	wood
1950	steel and wood
1970	aluminium and wood
2015	glass reinforced plastic (GRP), polyester

Table 2 shows information about some of the materials used to make the body of caravans.

Material		Density (g/cm ³)	Stiffness (GPa)	Melting Point (°C)	Strength (MPa)	Is it brittle?	Does it corrode?
wood		0.6			1 000	No	no but rots when wet
steel		7.8	210	1 357	1 200	No	yes producing rust
aluminium		2.7	69	660	90	No	yes producing a resistant coating
GRP	glass fibres	2.4	180	1 400	3 500	Yes	No
	epoxy resin	1.5			60	Yes	No
polyester		1.9	150	121	250	Yes	No

Use the information above to answer the following questions.

- (a) Describe how the weight and strength of caravan bodies changed between 1920 and 1970. [4]

Weight:

.....

Strength:

.....



- (b) (i) State **two** advantages of the materials used in 2015 over those used previously. [2]

1.

2.

- (ii) State **one** disadvantage of the materials used in 2015 over those used previously. [1]

.....

- (c) GRP is a common composite material, in which glass fibres are mixed with epoxy resin.

- (i) A GRP panel contains a volume of 300 cm^3 of epoxy resin.

Use the equation: [3]

$$\text{mass} = \text{density} \times \text{volume}$$

to calculate the mass of the resin.

mass = g

- (ii) I. State the strength of glass fibre in N/cm^2 . [2]

$$(1 \text{ MPa} = 100 \text{ N/cm}^2)$$

strength = N/cm^2

- II. The cross-sectional area (csa) of a glass fibre is 0.00015 cm^2 .

Use your answer in part (c)(ii)I. and the equation:

$$\text{force} = \text{strength} \times \text{csa}$$

to calculate the force required to break it. [2]

force = N

END OF PAPER





THE PERIODIC TABLE

Group

1

2

3

4

5

6

7

0

1 H Hydrogen 1 4 He Helium 2																											
7 Li Lithium 3		9 Be Beryllium 4																									
23 Na Sodium 11		24 Mg Magnesium 12																									
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30														
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	127 I Iodine 53	131 Xe Xenon 54									
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86								
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																							
Key																											

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

16

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